

# The SageLang Programming Language

## A Comprehensive Guide

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# The SageLang Programming Language: A Comprehensive Guide

## Executive Summary

**SageLang** is a **Python-inspired, systems-oriented programming language** written in C. It combines familiar Python syntax (indentation-based blocks, dynamic typing) with low-level systems capabilities (garbage collection, exception handling, generators, and module imports). The language now supports ten execution backends (C, LLVM IR, native assembly, bytecode VM, SageMetal VM, JIT, AOT, Kotlin/Android) and a self-hosted interpreter written in Sage itself. As of v3, Sage features structural value equality in uniqueness checks, safe non-hanging string/value repeating, and robust tab/whitespace token checks in sandbox security guards. This guide documents the language design, internal architecture, runtime semantics, and practical usage patterns derived from the complete C source implementation.

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## Part 1: Language Overview and Design Philosophy

### 1.1 Design Goals and Target Use Case

SageLang is designed as an **educational and practical embedded scripting language** that:

- **Bridges Python and C:** Provides Python-like syntax and dynamic typing while running in a minimal C footprint suitable for the RP2040.

- **Supports Systems Programming:** Unlike pure Python, SageLang includes explicit garbage collection control, class-based OOP, and low-level value representation suitable for embedded contexts.
- **Phases Incrementally:** The language grows through discrete, completed phases (1–18), each adding a cohesive feature set without breaking prior functionality.
- **Prioritizes Correctness:** Uses explicit memory management (via mark-and-sweep GC), scoped environments, and exception handling rather than relying on OS-level resource management.

## 1.2 Language Characteristics

| Feature                 | Details                                                                                                                                                                                                                              |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Syntax</b>           | Python-like: indentation-based blocks, <code>:</code> terminators, keywords like <code>let</code> , <code>var</code> , <code>proc</code> , <code>class</code>                                                                        |
| <b>Type System</b>      | Dynamically typed; values carry runtime type tags (numbers, strings, bools, arrays, dicts, tuples, classes, generators)                                                                                                              |
| <b>Scoping</b>          | Lexical scoping via nested environments; each block/function/class creates a child environment                                                                                                                                       |
| <b>Memory</b>           | Mark-and-sweep (tracing), ARC (reference counting), and ORC (optimized reference counting with cycle detection) modes                                                                                                                |
| <b>OOP</b>              | Class-based inheritance with single parent, methods, instance fields, <code>self</code> parameter, <code>super.init()</code> auto-self                                                                                               |
| <b>Control Flow</b>     | <code>if/else</code> , <code>while</code> , <code>for...in</code> , <code>break</code> , <code>continue</code> , <code>return</code> , <code>try/catch/finally</code> , <code>raise</code> , <code>yield</code> , <code>defer</code> |
| <b>Data Structures</b>  | Arrays (dynamic), dicts (string-keyed), tuples (immutable), slicing, indexing                                                                                                                                                        |
| <b>Functions</b>        | First-class <code>proc</code> declarations with closures; anonymous <code>proc</code> expressions ( <code>proc(x): body end</code> ); native C functions                                                                             |
| <b>Generators</b>       | Full <code>yield</code> support with resumable state; <code>next()</code> function to iterate                                                                                                                                        |
| <b>Exceptions</b>       | <code>try/catch/finally/raise</code> with explicit exception objects and message strings                                                                                                                                             |
| <b>Modules</b>          | <code>import module</code> , <code>import module as alias</code> , <code>from module import x</code> , <code>y</code> , <code>from module import x as y</code>                                                                       |
| <b>Standard Library</b> | Native modules: <code>math</code> , <code>io</code> , <code>string</code> , <code>sys</code> , <code>thread</code> , <code>fat</code> , <code>socket</code> , <code>tcp</code> , <code>http</code> , <code>ssl</code>                |
| <b>Networking</b>       | POSIX sockets, TCP, HTTP/HTTPS (libcurl), SSL/TLS (OpenSSL)                                                                                                                                                                          |
| <b>Concurrency</b>      | Multi-threaded <code>proc</code> , <code>async/await</code> , <code>atomics</code> , <code>semaphores</code> , <code>condvars</code> , <code>rwlocks</code> , SMP detection                                                          |
| <b>GPU Graphics</b>     | Vulkan + OpenGL 4.5 backends, handle-based API, 100+ functions, Android support                                                                                                                                                      |
| <b>UI Widgets</b>       | Immediate-mode GUI: windows, panels, buttons, sliders, menus, text inputs                                                                                                                                                            |
| <b>Compilation</b>      | C backend, LLVM IR (with GPU support), native assembly (x86-64, aarch64, rv64, mips), JIT self-extracting executables (with module bundling), plus initial profile suffixes for bare-metal / OSdev / UEFI targets                    |

LLVM codegen has an additional import optimization path: both the C LLVM backend and the self-hosted LLVM backend resolve `from module import CONST` for foldable top-level constants during code generation (including `as` aliases).

### 1.3 Execution Model

SageLang uses a shared front-end with multiple execution backends:

1. **Source Code** → **Lexer** (tokenization with indentation tracking)
2. **Tokens** → **Parser** (recursive descent, builds AST)
3. **AST** → one of five backends:
  - **AST interpreter** (tree-walking, default)
  - **Bytecode compiler + VM** (stack-based, faster for hot loops)
  - **SGVM binary** (`--sgvm`, binary bytecode artifact)
  - **Lily Transpiler** (`--compile-to-lily`, `--compile-from-lily`)
  - **C codegen** (`--emit-c` / `--compile`)
  - **LLVM IR** (`--emit-llvm` / `--compile-llvm`, with GPU support)
  - **Native assembly** (`--emit-asm` / `--compile-native`, x86-64/aarch64/rv64/mips)
  - **Freestanding ELF** (`--compile-bare`, bare-metal kernel output)
  - **UEFI PE** (`--compile-uefi`, EFI application output)
  - **SageMetal VM** (`make metal-vm`, freestanding bytecode object)
4. **Runtime Execution**: Execution by the MetalVM engine integrated into `sage` or via standalone `sgvm` tool.
5. **Runtime Values** stored in the shared **heap** managed by **GC**

All execution modes share the same object model: a **global environment**, nested **child environments** for scopes, and **tagged Value objects** that are either immediate (numbers, bools) or GC-managed heap values (arrays, dicts, strings, classes, instances, functions, generators).

For LLVM-compiled workloads, `from module import CONST` is resolved at compile time for foldable top-level module constants (numbers/strings/bools/nil plus simple constant expressions). In the C LLVM backend, GPU module constants are also resolved at compile time and GPU calls emit direct bridges to the pure C GPU API (`sgpu_*` in `gpu_api.h`), supporting both Vulkan and OpenGL backends. The bytecode VM provides 30 dedicated GPU opcodes for frame-loop hot paths.

The bytecode VM operates in hybrid mode by default: expressions, variables, loops (including break/continue), and function calls compile to stack bytecode, while unsupported constructs (classes, imports, exceptions, generators) fall back to the AST interpreter via `BC_OP_EXEC_AST_STMT`. This gives measurable speedups on loop-heavy workloads while maintaining full language coverage. Use `sage --runtime bytecode` or `sage --runtime auto` to enable.

### 1.4 Performance Characteristics

Sage ships with a benchmark suite (`benchmarks/01_fibonacci.sage` through `10_primes_sieve.sage`) with paired Python 3 implementations. Run `make benchmark-python` to compare. Typical results on the same workloads:

- **LLVM compiled**: 2-8x faster than CPython (fibonacci 3.9x, loop sum 7.7x)
  - **C compiled**: 2-4x faster than CPython on most workloads
  - **AST interpreter**: on par with CPython (faster on string/array ops, slower on recursion)
  - **Bytecode VM**: similar to AST, faster on tight loops due to reduced dispatch overhead
-

## Part 2: Internal Architecture and Core Modules

### 2.1 Module Dependency Graph

```
main.c
|- lexer.c / lexer.h      [Tokenization, indentation tracking]
|- parser.c / parser.h    [AST construction via recursive descent]
|- interpreter.c / interpreter.h [Tree-walking evaluation]
|- value.c / value.h      [Runtime value representation]
|- env.c / env.h          [Lexical scoping via linked-list environments]
|- gc.c / gc.h            [Mark-and-sweep garbage collection]
|- ast.c / ast.h          [AST node factory functions]
|- token.h                [Token type enumeration]
|- module.c / module.h    [Module loading, caching, imports]
|- compiler.c             [C code generation backend]
|- llvm_backend.c         [LLVM IR generation (with GPU support)]
|- llvm_runtime.c         [Standalone runtime for LLVM-compiled programs]
|- codegen.c / codegen.h  [Native assembly (x86-64, aarch64, rv64, mips)]
|- graphics.c / graphics.h [Vulkan GPU module for interpreter]
|- gpu_api.c / gpu_api.h  [Pure C GPU API (Vulkan + OpenGL)]
`- src/vm/                [Bytecode VM: bytecode.c, vm.c, program.c, runtime.c]
```

### 2.2 Lexer (lexer.c / lexer.h)

**Responsibility:** Convert source text into a stream of **tokens** while tracking indentation levels (Python-style).

**Key Data Structures:** - **Token:** { TokenType type, const char\* start, int length, int line }  
- **Lexer State:** { const char\* start, current, int line, int at\_beginning\_of\_line } - **Indent Stack:** Array tracking nesting depth; generates TOKEN\_INDENT / TOKEN\_DEDENT automatically

**Token Types** (from token.h): - **Keywords:** and, as, async, await, break, case, catch, class, comptime, continue, default, defer, elif, else, end, enum, false, finally, for, from, if, import, in, init, let, macro, match, nil, not, or, print, proc, quote, raise, return, self, struct, super, trait, true, try, unquote, unsafe, var, while, yield, @ - *Note: print, end, match, init, enum, struct, and trait are soft keywords and can also be used as variable, property, or method names.* - **Operators:** +, -, \*, /, =, ==, !=, <, >, <=, >=, and, or, &, |, ^, ~, <<, >> - **Punctuation:** (, ), [, ], {, }, :, ,, . - **Structural:** INDENT, DEDENT, NEWLINE, DOC\_COMMENT, EOF, ERROR

**Indentation Handling** (crucial for Python-like syntax):

**Resource Management:** - **SAGE\_MAX\_READ\_SIZE:** A 100MB limit is enforced on all file I/O and network receive operations to prevent memory exhaustion DoS attacks.

- Tracks column position at the start of each line.
- Compares current indent to stack top:
  - **Increase:** Push new level, emit TOKEN\_INDENT.
  - **Decrease:** Pop levels, emit TOKEN\_DEDENT for each popped level.
  - **Mismatch:** Error if indent doesn't align with a previous level.

**Example:**

```
proc greet(name):
  let msg = "Hello, " + name
  print msg
# Tokens: proc, identifier(greet), (, identifier(name), ), :, newline,
#         indent, let, identifier(msg), =, string, newline,
#         print, identifier(msg), newline,
#         dedent, eof
```

## 2.3 Parser (parser.c)

**Responsibility:** Build an **Abstract Syntax Tree (AST)** from tokens using **recursive descent parsing**.

**Parsing Levels** (operator precedence):

```
expression    → assignment
assignment    → logical_or ("=" assignment)?
logical_or    → logical_and ("or" logical_and)*
logical_and   → equality ("and" equality)*
equality      → comparison (("==" | "!=") comparison)*
comparison    → addition (<"<" | ">" | "<=" | ">=") addition)*
addition      → term (("+" | "-") term)*
term          → unary (("*" | "/" ) unary)*
unary         → ("-" unary) | postfix
postfix       → primary ("[" ... "]" | "." property)*
primary       → NUMBER | STRING | BOOL | NIL | "(" expr ")" | "[" array "]" | "{" dict "}" | IDENTIFIER
```

**Statement Parsing** (top-down recursive descent):

```
statement     → if_stmt | while_stmt | for_stmt | return_stmt | break | continue | try_stmt | raise_stmt
declaration   → class_decl | proc_decl | let_decl | statement
```

**Key AST Node Types** (see ast.h/ast.c): - **Expr** variants: `EXPR_NUMBER`, `EXPR_STRING`, `EXPR_BOOL`, `EXPR_NIL`, `EXPR_VARIABLE`, `EXPR_CALL`, `EXPR_BINARY`, `EXPR_UNARY`, `EXPR_INDEX`, `EXPR_SLICE`, `EXPR_ARRAY`, `EXPR_DICT`, `EXPR_TUPLE`, `EXPR_GET`, `EXPR_SET` - **Stmt** variants: `STMT_EXPRESSION`, `STMT_LET`, `STMT_PRINT`, `STMT_IF`, `STMT_WHILE`, `STMT_FOR`, `STMT_BLOCK`, `STMT_PROC`, `STMT_CLASS`, `STMT_RETURN`, `STMT_BREAK`, `STMT_CONTINUE`, `STMT_TRY`, `STMT_RAISE`, `STMT_YIELD`, `STMT_DEFER`, `STMT_MATCH`, `STMT_IMPORT`

**Example Parsing:**

Input: `let x = 10 + 5`

```
Output: Stmt {
  type: STMT_LET,
  name: Token("x"),
  initializer: Expr {
    type: EXPR_BINARY,
    left: Expr { type: EXPR_NUMBER, value: 10 },
    op: Token(PLUS),
    right: Expr { type: EXPR_NUMBER, value: 5 }
  }
}
```

## 2.4 Value System (value.h / value.c)

**Responsibility:** Define and manage **runtime value types** and their operations.

**Core Value Type:**

```
struct Value {
  ValueType type; // VAL_NUMBER, VAL_STRING, VAL_ARRAY, etc.
  union {
    double number;
    int boolean;
    char* string;
    NativeFn native;
    FunctionValue* function;
    ArrayValue* array;
    DictValue* dict;
  };
}
```

```

    TupleValue* tuple;
    ClassValue* class_val;
    InstanceValue* instance;
    ExceptionValue* exception;
    GeneratorValue* generator;
} as;
};

```

### Value Types:

| Type          | Storage                      | Mutable | Use Case                                     |
|---------------|------------------------------|---------|----------------------------------------------|
| VAL_NUMBER    | double (on stack)            | Yes     | Integers, floats; 10, 3.14                   |
| VAL_BOOL      | int (on stack)               | Yes     | Boolean logic; true, false                   |
| VAL_NIL       | N/A                          | No      | Null value; nil                              |
| VAL_STRING    | char* (heap)                 | No      | Text; immutable (assign new to mutate)       |
| VAL_ARRAY     | ArrayValue* (dynamic)        | Yes     | Lists; [1, 2, 3], push(), indexing           |
| VAL_DICT      | DictValue* (string-keyed)    | Yes     | Maps; {"key": value}, dict_set()             |
| VAL_TUPLE     | TupleValue* (immutable)      | No      | Fixed-size sequences; (1, 2, 3)              |
| VAL_FUNCTION  | FunctionValue* (closure)     | No      | User-defined procs with captured environment |
| VAL_NATIVE    | NativeFn (C function ptr)    | No      | Built-in functions like len(), print()       |
| VAL_CLASS     | ClassValue* (metadata)       | No      | Class definition with methods and parent     |
| VAL_INSTANCE  | InstanceValue* (fields dict) | Yes     | Object; instance of a class                  |
| VAL_EXCEPTION | ExceptionValue* (message)    | No      | Exception; raised via raise                  |
| VAL_GENERATOR | GeneratorValue* (resumable)  | Yes     | Iterable; maintains state across yield       |

**Array Operations:** - array\_push(arr, val): Append to dynamic array (resizes if needed). - array\_get(arr, index), array\_set(arr, index, val): Access/modify. - array\_slice(arr, start, end): Subarray (negative indices supported).

**Dictionary Operations** (O(1) amortized via hash table): - dict\_set(dict, key, value): Insert/update using FNV-1a hashing with open-addressing and linear probing. Table grows at 75% load factor. - dict\_get(dict, key), dict\_has(dict, key): O(1) average lookup. - dict\_delete(dict, key): Backward-shift deletion preserves probe chain integrity (no tombstones). - dict\_keys(dict), dict\_values(dict): Return as arrays (iterates capacity, skips empty slots).

**String Operations:** - string\_split(str, delim), string\_join(arr, sep): Splitting/joining arrays. - string\_replace(str, old, new), string\_upper/lower/strip(str): Transformation.

**Class/Instance Operations:** - class\_create(name, parent): Define a class with optional parent. - class\_add\_method(class, name, method\_stmt): Add methods. - instance\_create(class\_def): Create instance with empty field dictionary. - instance\_set\_field(), instance\_get\_field(): Access instance variables.

## 2.5 Environment and Scoping (env.h / env.c)

**Responsibility:** Implement **lexical scoping** via a chain of nested environments (linked list).

**Environment Structure:**

```
struct Env {
    EnvNode* head;           // Variables defined in this scope (linked list)
    struct Env* parent;      // Enclosing scope
    struct Env* alloc_next;  // Internal registry for GC/shutdown cleanup
    int marked;              // GC mark flag (0 = unmarked, 1 = reachable)
};

struct EnvNode {
    char* name;              // Variable name
    Value value;             // Stored value
    struct EnvNode* next;
};
```

**Scoping Rules:** - **env\_define(env, name, length, value)**: Create or update a variable **in the current scope only**. If variable exists in current scope, update it; otherwise, create new. - **env\_get(env, name, length, value\*)**: Look up a variable, searching current scope then parent chains recursively. Return 1 if found, 0 if not. - **env\_assign(env, name, length, value)**: Assign to an **existing** variable by searching up the scope chain. Used for reassigning already-defined variables.

**Example Scoping:**

```
Global: { x: 10, y: 20 }
  Block: { z: 30 } → parent: Global
    Inner: { x: 100 } → parent: Block
```

In Inner scope:

- env\_get("x") → 100 (found in current scope)
- env\_get("z") → 30 (found in parent Block)
- env\_get("y") → 20 (found in Global via parent chain)

## 2.6 Garbage Collector (gc.h / gc.c)

**Responsibility:** Automatically reclaim heap memory using tracing or reference counting.

**GC Modes:**

- **Tracing** (--gc:tracing, default): Concurrent tri-color mark-sweep with SATB write barriers. Best for general use with sub-millisecond pauses.
- **ARC** (--gc:arc): Deterministic reference counting. High performance for linear object graphs; does not collect cycles.
- **ORC** (--gc:orc): Optimized Reference Counting with Lins' trial deletion cycle collector. Combines the determinism of ARC with a robust cycle detection algorithm for complex graphs.

**Tracing GC Overview:** - **Mark Phase:** Starting from **root set** (global environment, function registry, call stack), mark all reachable objects. Born-black objects during concurrent marking. - **Sweep Phase:** Free all unmarked objects and add them to a free pool. - **Triggering:** Automatic based on object count and heap pressure, or manual via gc\_collect().

**GC Configuration:**

```
#define GC_HEAP_SIZE (1024 * 1024)  // 1 MB theoretical heap
#define GC_THRESHOLD 1000           // Collect after 1000 object allocations
```

**GC Header** (prepended to all heap allocations):

```

struct GCHeader {
    int marked;      // 1 = marked (reachable), 0 = unmarked (candidate for freeing)
    int type;        // Object type (VAL_ARRAY, VAL_DICT, etc.)
    void* next;      // Linked list of all allocated objects
};

```

**Root Set:** - Global environment (`g_global_env`) and its entire scope chain. - Function registry (linked list of all defined proc declarations). - Call stack (environments of active function calls).

**GC Statistics** (via `gc_stats()` native function):

```

bytes_allocated: Total heap allocated (cumulative)
num_objects: Current number of live objects
collections: Number of GC runs so far
objects_freed: Objects freed in most recent collection
next_gc: Objects until next collection threshold

```

**Usage:**

```

# Automatic collection when threshold hit
let arr = [1, 2, 3] # Allocates array

# Manual trigger
gc_collect()

# Check stats
let stats = gc_stats()
print stats
# Output: {"bytes_allocated": 1024, "num_objects": 5, "collections": 2, ...}

# Control GC
gc_disable() # Pause automatic collection
gc_enable()  # Resume automatic collection

```

## 2.7 Interpreter (interpreter.c / interpreter.h)

**Responsibility:** Tree-walking evaluation of the AST, maintaining runtime state and executing statements/expressions.

**Execution Result** (tracks control flow):

```

struct ExecResult {
    Value value;          // Return value
    int is_returning;     // `return` statement hit
    int is_breaking;      // `break` statement hit
    int is_continuing;    // `continue` statement hit
    int is_throwing;      // Exception raised
    Value exception_value; // Exception object
    int is_yielding;      // `yield` hit (generators)
    void* next_stmt;      // Resume point for generators
};

```

**Statement Execution** (`interpret(stmt, env)`): - **STMT\_LET**: Define variable (`let` or `var`) in current scope; evaluate initializer if present. - **STMT\_EXPRESSION**: Evaluate expression for side effects. - **STMT\_PRINT**: Evaluate expression and print value. - **STMT\_IF**: Evaluate condition; execute then-branch or else-branch. - **STMT\_WHILE**: Loop while condition is truthy; respect `break`/`continue`. - **STMT\_FOR**: Iterate over array/range; update loop variable in scope. - **STMT\_BLOCK**: Create child scope, execute statements in sequence.

- **STMT\_PROC**: Register function in function registry (not executed). - **STMT\_CLASS**: Create class object; register methods. - **STMT\_RETURN**: Exit function with value. - **STMT\_TRY**: Execute try-block; on exception, search matching catch-clause; always run finally-block. - **STMT\_RAISE**: Throw exception (string or exception object). - **STMT\_YIELD**: In generators, pause execution and return value; next `next()` call resumes. - **STMT\_IMPORT**: Load module; populate current environment with imports.

**Expression Evaluation** (`eval_expr(expr, env)`): - **Literals**: Numbers, strings, bools, nil returned as-is. - **Variables**: Look up in environment via `env_get()`. - **Binary Ops**: Evaluate left, then right (short-circuit for `and/or`); apply operator. - Arithmetic: `+`, `-`, `*`, `/` on numbers; `+` on strings (concatenation). - Comparison: `<`, `>`, `<=`, `>=` on numbers; `==`, `!=` on any type. - Logical: `and`, `or` with short-circuit evaluation. - **Calls**: Look up function (user-defined or native); bind arguments to parameters; execute in new environment; return result. - **Indexing**: Evaluate array and index; return element or slice if range. - **Array/Dict/Tuple Construction**: Evaluate elements; construct heap-allocated structure. - **Property Access**: For instances, look up field in instance's field dict.

**Function Call Mechanics**: 1. Resolve callee to `ProcStmt*` or `NativeFn`. 2. Create **child environment** with current scope as parent. 3. Bind parameters to arguments via `env_define()`. 4. Execute function body statements. 5. On **return**, break with value; otherwise use `nil`. 6. Return to caller with result.

**Generator Execution** (Phase 7): 1. `proc my_gen()`: creates a function that yields values. 2. Call `my_gen()` → returns `VAL_GENERATOR` value. 3. Call `next(gen)` → initializes generator environment, runs until first yield. 4. Each `next()` resumes from saved `current_stmt`; yields next value or `nil` if exhausted.

**Native Function Registry**: SageLang provides built-in functions injected into global environment via `init_stdlib(env)`:

| Function                               | Signature                                      | Purpose                                                                               |
|----------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------|
| <code>len(arr/str/dict)</code>         | <code>Value → number</code>                    | Get length of collection                                                              |
| <code>push(arr, val)</code>            | <code>(array, value) → nil</code>              | Append to array                                                                       |
| <code>pop(arr)</code>                  | <code>array → value</code>                     | Remove and return last element                                                        |
| <code>range(start, end)</code>         | <code>(number, number) → array</code>          | Generate list [ <code>start</code> , <code>start+1</code> , ..., <code>end-1</code> ] |
| <code>slice(arr, start, end)</code>    | <code>(array, number, number) → array</code>   | Extract subarray                                                                      |
| <code>split(str, delim)</code>         | <code>(string, string) → array</code>          | Split string by delimiter                                                             |
| <code>join(arr, sep)</code>            | <code>(array, string) → string</code>          | Join array elements with separator                                                    |
| <code>replace(str, old, new)</code>    | <code>(string, string, string) → string</code> | Replace all occurrences                                                               |
| <code>upper(str)</code>                | <code>string → string</code>                   | Uppercase                                                                             |
| <code>lower(str)</code>                | <code>string → string</code>                   | Lowercase                                                                             |
| <code>strip(str)</code>                | <code>string → string</code>                   | Trim whitespace                                                                       |
| <code>str(val)</code>                  | <code>value → string</code>                    | Convert to string                                                                     |
| <code>tonumber(str)</code>             | <code>string → number</code>                   | Parse number                                                                          |
| <code>input()</code>                   | <code>() → string</code>                       | Read line from stdin                                                                  |
| <code>clock()</code>                   | <code>() → number</code>                       | Elapsed seconds since process start                                                   |
| <code>dict_keys(dict)</code>           | <code>dict → array</code>                      | Get keys as array                                                                     |
| <code>dict_values(dict)</code>         | <code>dict → array</code>                      | Get values as array                                                                   |
| <code>dict_has(dict, key)</code>       | <code>(dict, string) → bool</code>             | Check if key exists                                                                   |
| <code>dict_delete(dict, key)</code>    | <code>(dict, string) → nil</code>              | Remove key from dict                                                                  |
| <code>gc_collect()</code>              | <code>() → nil</code>                          | Trigger garbage collection                                                            |
| <code>gc_stats()</code>                | <code>() → dict</code>                         | Get GC statistics                                                                     |
| <code>gc_enable(), gc_disable()</code> | <code>() → nil</code>                          | Control automatic GC                                                                  |

| Function                                     | Signature                                           | Purpose                                     |
|----------------------------------------------|-----------------------------------------------------|---------------------------------------------|
| <code>next(gen)</code>                       | <code>generator → value</code>                      | Resume generator, get next yielded value    |
| <code>ffi_open(path)</code>                  | <code>string → clib</code>                          | Open shared library via <code>dlopen</code> |
| <code>ffi_call(lib, func, ret, ...)</code>   | <code>(clib, string, string, ...) → value</code>    | Call C function in library                  |
| <code>ffi_sym(lib, name)</code>              | <code>(clib, string) → bool</code>                  | Check if symbol exists in library           |
| <code>ffi_close(lib)</code>                  | <code>clib → nil</code>                             | Close shared library handle                 |
| <code>mem_alloc(size)</code>                 | <code>number → pointer</code>                       | Allocate raw memory (zero-initialized)      |
| <code>mem_free(ptr)</code>                   | <code>pointer → nil</code>                          | Free allocated memory                       |
| <code>mem_read(ptr, off, type)</code>        | <code>(pointer, number, string) → value</code>      | Read value at <code>ptr+offset</code>       |
| <code>mem_write(ptr, off, type, val)</code>  | <code>(pointer, number, string, value) → nil</code> | Write value at <code>ptr+offset</code>      |
| <code>mem_size(ptr)</code>                   | <code>pointer → number</code>                       | Get allocation size                         |
| <code>addressof(val)</code>                  | <code>value → number</code>                         | Get memory address of a value               |
| <code>asm_exec(code, ret, ...)</code>        | <code>(string, string, ...) → value</code>          | Compile and execute assembly                |
| <code>asm_compile(code, arch, out)</code>    | <code>(string, string, string) → bool</code>        | Cross-compile assembly to object file       |
| <code>asm_arch()</code>                      | <code>() → string</code>                            | Get host architecture name                  |
| <code>struct_def(fields)</code>              | <code>array → dict</code>                           | Define C struct layout with alignment       |
| <code>struct_new(def)</code>                 | <code>dict → pointer</code>                         | Allocate zeroed struct instance             |
| <code>struct_get(ptr, def, name)</code>      | <code>(pointer, dict, string) → value</code>        | Read struct field                           |
| <code>struct_set(ptr, def, name, val)</code> | <code>(pointer, dict, string, value) → nil</code>   | Write struct field                          |
| <code>struct_size(def)</code>                | <code>dict → number</code>                          | Get total struct size                       |
| <code>cpu_count()</code>                     | <code>() → number</code>                            | Logical CPU core count                      |
| <code>cpu_physical_cores()</code>            | <code>() → number</code>                            | Physical CPU core count                     |
| <code>cpu_has_hyperthreading()</code>        | <code>() → bool</code>                              | HT check                                    |
| <code>thread_set_affinity(id)</code>         | <code>number → number</code>                        | Pin current thread to core                  |
| <code>thread_get_core()</code>               | <code>() → number</code>                            | Current CPU core ID                         |
| <code>atomic_new(val)</code>                 | <code>value → pointer</code>                        | Create atomic long                          |
| <code>atomic_load(a)</code>                  | <code>pointer → number</code>                       | Atomic read                                 |
| <code>atomic_store(a, v)</code>              | <code>(pointer, number) → nil</code>                | Atomic write                                |
| <code>atomic_add(a, v)</code>                | <code>(pointer, number) → number</code>             | Atomic fetch-and-add                        |
| <code>atomic_cas(a, e, d)</code>             | <code>(pointer, num, num) → bool</code>             | Atomic compare-and-swap                     |
| <code>atomic_exchange(a, v)</code>           | <code>(pointer, number) → number</code>             | Atomic fetch-and-exchange                   |
| <code>sem_new(n)</code>                      | <code>number → pointer</code>                       | Create POSIX semaphore                      |
| <code>sem_wait(s)</code>                     | <code>pointer → nil</code>                          | Blocking wait                               |
| <code>sem_post(s)</code>                     | <code>pointer → nil</code>                          | Signal/release                              |
| <code>sem_trywait(s)</code>                  | <code>pointer → bool</code>                         | Non-blocking wait                           |

## 2.8 Module System (`module.h` / `module.c`)

**Responsibility:** Load `.sage` files as reusable modules and manage imports.

**Module Structure:**

```

struct Module {
    char* name;           // Module identifier (e.g., "math")
    char* path;           // Full filesystem path to .sage file
    Environment* env;     // Module's own environment (exports)
    bool is_loaded;       // Execution complete?
    bool is_loading;      // Currently executing? (cycle detection)
    struct Module* next;  // For linked-list cache
};

struct ModuleCache {
    Module* modules;      // All loaded modules
    char** search_paths;  // Directories to search (., ./lib, ./modules)
    int search_path_count;
};

```

**Module Lifecycle:** 1. `load_module(cache, "math")`: Search in `search_paths` for `math.sage` or `math/__init__.sage`; create `Module` struct; add to cache. 2. `execute_module(module, global_env)`: Read file; lex/parse/interpret in module's own environment (child of global). 3. **Exports**: Any top-level proc, class, let in module becomes available for import.

**Import Statements** (parsed in `parser.c`, executed in `interpreter.c`):

```

# Import entire module (unaliased)
import math
# → Available as: math (currently placeholder; future: module.sin(), etc.)

# Import module with alias
import math as m
# → Available as: m

# Import specific items
from math import sin, cos
# → sin and cos directly available in scope

# Import specific items with aliases
from math import sin as sine, cos as cosine
# → sine, cosine available in scope

# Star import (all exports)
from math import *
# → All module exports available directly

```

**Search Path Resolution:** 1. Try `{search_path}/module_name.sage`. 2. Try `{search_path}/module_name/__init__.sage`. 3. If not found in any search path, error.

**Default Search Paths:**

```

- "."           (current directory)
- "./lib"       (local lib subdirectory)
- "./modules"   (local modules subdirectory)

```

**Import Processing:** - `import_all(env, "math")`: Load module, define module name in env. - `import_from(env, "math", items[], count)`: Load module, look up items, define in env. - `import_as(env, "math", "m")`: Load module, define alias in env.

**Compiled LLVM Constant Imports:** - Applies to both LLVM pipelines: the C backend (`src/c/llvm_backend.c`) and the self-hosted backend (`src/sage/llvm_backend.sage`). - `from module import NAME` (and `from module import NAME as ALIAS`) is pre-resolved from module top-level `let`

constants when they are compile-time foldable. - Supported imported value shapes are scalar constants (number, string, bool, nil) and expressions composed from previously resolved constants. - Search paths for this compile-time resolution match module lookup defaults: ./, ./lib, and ./modules. - In the C LLVM backend, from gpu import CONST is also resolved from the built-in GPU constant table. - Unresolved imported constants are treated as compile errors in LLVM codegen rather than generating unresolved %NAME loads.

**Circular Dependency Detection:** is\_loading flag prevents infinite loops during module execution.

---

## Part 3: Language Features by Phase

### Phase 1–3: Core Language (Arithmetic, Variables, Control Flow)

**Implemented:** Basic arithmetic, variable declaration, conditionals, loops, functions.

**Example:**

```
let x = 10
let y = 20
print x + y # Output: 30

if x > 5:
    print "x is greater than 5"
else:
    print "x is not greater than 5"

proc add(a, b):
    return a + b

print add(3, 4) # Output: 7
```

### Phase 4: Garbage Collection

**Added:** Mark-and-sweep GC automatically manages heap.

**Example:**

```
let arr = [1, 2, 3, 4, 5]
print len(arr) # 5

# GC runs automatically when threshold hit
gc_collect() # Force collection
let stats = gc_stats()
print stats["collections"] # Number of GC runs
```

### Phase 5–6: Data Structures and OOP

**Phase 5:** Arrays, dicts, tuples, slicing, indexing.

```
let arr = [1, 2, 3, 4, 5]
print arr[0] # 1
print arr[1:3] # [2, 3]
arr[0] = 10
print arr # [10, 2, 3, 4, 5]

let dict = {"name": "Alice", "age": 30}
print dict["name"] # "Alice"
```

```
dict["age"] = 31

let tuple = (1, 2, 3)
print tuple          # (1, 2, 3)  # Immutable
```

**Phase 6:** Classes, inheritance, instance methods.

```
class Animal:
    proc init(name):
        self.name = name
        self.age = 0

    proc birthday():
        self.age = self.age + 1

    proc speak():
        print self.name + " makes a sound"

class Dog(Animal):
    proc speak():
        print self.name + " barks"

let dog = Dog("Buddy")
print dog.name          # "Buddy"
dog.birthday()
print dog.age           # 1
dog.speak()             # "Buddy barks"
```

**Phase 7:** Advanced Control Flow, Exceptions, Generators

**Exceptions:**

```
proc divide(a, b):
    if b == 0:
        raise "Division by zero"
    return a / b

try:
    print divide(10, 0)
catch e:
    print "Caught error: " + e
finally:
    print "Cleanup done"
```

**Generators:**

```
proc count_up(n):
    let i = 0
    while i < n:
        yield i
        i = i + 1

let gen = count_up(3)
print next(gen)      # 0
print next(gen)      # 1
print next(gen)      # 2
print next(gen)      # nil (exhausted)
```

**Deferred Code** (Phase 7 addition): `defer` executes a statement on scope exit, in LIFO (Last-In, First-Out) order, which is extremely useful for cleanup:

```
proc read_and_print():
  let f = file_open("data.txt")
  defer file_close(f)
  print file_read(f)
```

**Pattern Matching** (Phase 7 addition): `match` allows matching a value against multiple patterns, with optional guards and a default fallback:

```
let x = 42
match x:
  case 1:
    print "one"
  case 42 if y > 0:
    print "forty-two with positive y"
  case 42:
    print "forty-two"
  default:
    print "other"
```

## Phase 8: Modules

```
# math.sage
proc sin(x):
  # Approximate sine
  return x # Stub

proc cos(x):
  return 1 # Stub

# main.sage
import math as m
print m.sin(0) # Calls math.sin in module's namespace

# Or:
from math import sin, cos
print sin(0)
print cos(0)
```

## Phase 9: Low-Level Programming

### Bitwise Operators:

```
# Bitwise AND, OR, XOR
print 5 & 3      # 1 (0101 & 0011 = 0001)
print 5 | 3      # 7 (0101 | 0011 = 0111)
print 5 ^ 3      # 6 (0101 ^ 0011 = 0110)

# Bitwise NOT
print ~0         # -1 (all bits flipped)

# Shift operators
print 1 << 4      # 16 (shift left by 4)
print 16 >> 2     # 4 (shift right by 2)
```

```
# Practical: extract lower nibble
let val = 255
let mask = 15
print val & mask    # 15
```

```
# Practical: check if bit is set
let x = 7
let bit2 = (x >> 2) & 1
print bit2 == 1    # true
```

**Bitwise Safety:** Shift amounts are validated at runtime — values outside 0-63 return 0 instead of causing C undefined behavior. Right-shift is arithmetic (sign-extending) on signed values. Floating-point operands are truncated to `long long` before bitwise operations.

**Unsafe Blocks:** As of version 4.0.1 (Specification 2.0), low-level memory operations or operations bypassing typical safety checks should be wrapped in an `unsafe` block. `unsafe` blocks must be terminated with the `end` keyword.

```
unsafe:
  # Memory primitives or FFI calls here
  let ptr = mem_alloc(16)
  mem_write(ptr, 0, "int", 42)
  mem_free(ptr)
end
```

### Foreign Function Interface (FFI):

```
# Open a shared C library
let libm = ffi_open("libm.so.6")

# Call C functions with return type and arguments
let result = ffi_call(libm, "sqrt", "double", 144.0)
print result    # 12

let s = ffi_call(libm, "sin", "double", 0.0)
print s        # 0

# Check if a symbol exists
print ffi_sym(libm, "cos")    # true
print ffi_sym(libm, "bogus")  # false

# Close when done
ffi_close(libm)

# Call libc functions
let libc = ffi_open("libc.so.6")
let n = ffi_call(libc, "strlen", "long", "hello")
print n    # 5
ffi_close(libc)
```

FFI supports return types: `"double"`, `"int"`, `"long"`, `"string"`, `"void"`, with up to 3 arguments (numeric or string). Passing more than 3 arguments returns an error. Library handles are tracked by the GC and properly freed on collection.

### Raw Memory Operations:

```
# Allocate 32 bytes of raw memory
let buf = mem_alloc(32)
```

```

print mem_size(buf)      # 32

# Write and read different types
mem_write(buf, 0, "byte", 42)
print mem_read(buf, 0, "byte")      # 42

mem_write(buf, 4, "int", 12345)
print mem_read(buf, 4, "int")      # 12345

mem_write(buf, 8, "double", 3.14)
print mem_read(buf, 8, "double")    # 3.14

# Get memory address of any value
let arr = [1, 2, 3]
print addressof(arr)    # memory address as number

# Free when done
mem_free(buf)

```

The `mem_read(ptr, offset, type)` function requires exactly 3 arguments, and `mem_write(ptr, offset, type, value)` requires exactly 4 arguments. Supported explicit type strings for `mem_read`/`mem_write` are: "byte" (1 byte), "int" (4 bytes), "double" (8 bytes), "string" (read-only, null-terminated). Allocations are capped at 64MB. Negative offsets are rejected. Bounds checking is enforced for owned memory (offset + type size must not exceed allocation). Double-free is prevented via handle nullification. Memory pointers are GC-tracked and freed on collection if owned.

**Inline Assembly** (x86-64, aarch64, rv64, mips):

```

# Detect host architecture
print asm_arch()      # "x86_64"

# Execute x86-64 assembly: return a constant
let val = asm_exec("    mov $42, %rax", "int")
print val              # 42

# Add two integers (args in rdi, rsi per System V ABI)
let sum = asm_exec("    mov %rdi, %rax\n    add %rsi, %rax", "int", 10, 32)
print sum              # 42

# Add two doubles (args in xmm0, xmm1)
let r = asm_exec("    addsd %xmm1, %xmm0", "double", 1.5, 2.7)
print r                # 4.2

# Cross-compile for aarch64 (requires aarch64-linux-gnu-as)
let ok = asm_compile("    mov x0, #42", "aarch64", "/tmp/out.o")

# Cross-compile for RISC-V 64 (requires riscv64-linux-gnu-as)
let ok2 = asm_compile("    li a0, 42", "rv64", "/tmp/out_rv.o")

```

Supported architectures: "x86\_64", "aarch64", "rv64", "mips". Return types: "int", "double", "void". Up to 4 numeric arguments.

**C Struct Interop:**

```

# Define a C-compatible struct: { int x; int y; double z; }
let Point = struct_def(["x", "int"], ["y", "int"], ["z", "double"]))
print struct_size(Point)    # 16 (with alignment)

```

```

# Allocate and populate
let p = struct_new(Point)
struct_set(p, Point, "x", 10)
struct_set(p, Point, "y", 20)
struct_set(p, Point, "z", 3.14)

# Read fields
print struct_get(p, Point, "x")    # 10
print struct_get(p, Point, "y")    # 20
print struct_get(p, Point, "z")    # 3.14

mem_free(p)

# Alignment example: { char a; double b; int c; }
# Layout: a@0, pad(7), b@8, c@16, pad(4) = 24 bytes
let S = struct_def(["a", "char"], ["b", "double"], ["c", "int"])
print struct_size(S)                # 24

Supported types: "char", "byte" (1), "short" (2), "int" (4), "long" (8), "float" (4), "double" (8),
"ptr" (8). Alignment follows C ABI rules.

```

---

## Part 4: Writing SageLang Programs

### 4.1 Basic Syntax Patterns

#### Variables and Types:

```

let x = 42                # Variable binding
var y = 100               # Explicitly mutable binding
let s = "Hello, World"   # String
let b = true              # Boolean
let arr = [1, 2, 3]       # Array
let dict = {"a": 1, "b": 2} # Dictionary
let tup = (1, "two", 3)   # Tuple (immutable)
let n = nil               # Nil (null)

```

#### Operators:

```

# Arithmetic
print 10 + 5    # 15
print 10 - 5    # 5
print 10 * 5    # 50
print 10 / 2    # 5
print -5        # -5

# Comparison
print 5 == 5    # true
print 5 != 3    # true
print 5 > 3     # true
print 5 < 10    # true

# Logical
print true and false # false
print true or false  # true

```

```
# Bitwise
print 5 & 3          # 1
print 5 | 3          # 7
print 5 ^ 3          # 6
print ~0             # -1
print 1 << 4          # 16
print 16 >> 2        # 4
```

### String Operations:

```
let a = "Hello"
let b = "World"
print a + " " + b    # "Hello World"

let words = "a,b,c".split(",")
print words           # ["a", "b", "c"]

let joined = ["x", "y"].join("-")
print joined          # "x-y"

print "hello".upper() # "HELLO"
print "HELLO".lower() # "hello"
```

### Arrays:

```
let arr = [10, 20, 30, 40, 50]
print len(arr)        # 5
print arr[0]          # 10
arr[0] = 100
print arr[0]          # 100
push(arr, 60)
print len(arr)        # 6

# Slicing
print arr[1:3]         # [20, 30]
print arr[2:]          # [30, 40, 50, 60] (from index 2 to end)
print arr[:2]          # [100, 20] (from start to index 2)
```

### Dictionaries:

```
let dict = {"name": "Alice", "age": 30}
print dict["name"]     # "Alice"
dict["age"] = 31
print dict_has(dict, "name") # true
print dict_keys(dict)   # ["name", "age"]
print dict_values(dict) # ["Alice", 31]

dict_delete(dict, "age")
print dict_has(dict, "age") # false
```

## 4.2 Control Flow Patterns

### Conditionals:

```
let x = 15

if x < 10:
```

```

    print "small"
else:
    print "large"

# Nested conditions
if x > 10:
    if x > 20:
        print "very large"
    else:
        print "medium"
else:
    print "small"

```

#### Loops:

```

# While loop
let i = 0
while i < 5:
    print i
    i = i + 1

# For loop over array
let arr = [10, 20, 30]
for item in arr:
    print item

# For loop over range
for i in range(0, 5):
    print i

# Break and continue
for i in range(0, 10):
    if i == 3:
        continue
    if i == 7:
        break
    print i

```

### 4.3 Functions and Closures

#### Function Definition and Call:

```

proc greet(name):
    print "Hello, " + name

greet("Alice")

proc add(a, b):
    return a + b

let result = add(5, 3)
print result          # 8

```

#### Closures (captured environment):

```

proc make_multiplier(factor):
    proc multiply(x):

```

```

        return x * factor      # Captures 'factor' from enclosing scope
    return multiply

let times3 = make_multiplier(3)
let times5 = make_multiplier(5)

print times3(10)              # 30
print times5(10)              # 50

```

#### 4.4 Structs, Enums, Traits, and Classes

**Struct Definition (Phase 1.7):** SageLang provides native `struct` support as a lightweight alternative to classes. Structs automatically generate `init`, `to_string`, and equality methods based on their fields. Under the hood, structs are represented as standard class/instance values (`VAL_CLASS/VAL_INSTANCE`).

```

struct Point:
    x
    y
end

let p = Point(10, 20)
print p.x      # 10
print p        # Point(x=10, y=20)

```

**Enum Definition (Phase 1.7):** Native C-like `enum` support maps variant names to sequential integers starting from 0. At runtime, enums are represented as dictionaries. Note that tagged unions (ADTs with associated data) are still implemented via the standard library (`std.enum`).

```

enum Color:
    Red
    Green
    Blue

print Color.Red    # 0
print Color.Green  # 1

```

**Trait Definition (Phase 1.7):** Traits define an interface contract of method signatures. A trait compiles down to a dictionary containing the list of required method names.

```

trait Printable:
    proc to_string(self)
end

```

#### Class Definition:

```

class Point:
    proc init(x, y):
        self.x = x
        self.y = y

    proc distance_from_origin():
        # Simplified (no sqrt)
        return self.x * self.x + self.y * self.y

let p = Point(3, 4)
print p.x      # 3
print p.y      # 4
print p.distance_from_origin() # 25

```

### Inheritance:

```
class Vehicle:
  proc init(wheels):
    self.wheels = wheels

  proc describe():
    print "Vehicle with " + str(self.wheels) + " wheels"

class Car(Vehicle):
  proc init(wheels, doors):
    self.wheels = wheels
    self.doors = doors

  proc describe():
    print "Car with " + str(self.wheels) + " wheels and " + str(self.doors) + " doors"

let car = Car(4, 4)
car.describe()           # "Car with 4 wheels and 4 doors"
```

### Calling Parent Methods with super:

Use `super.init(args)` to call the parent class constructor, and `super.method(args)` to call any parent method. `self` is automatically injected. This works with chained inheritance (3+ levels, e.g.,  $A \rightarrow B \rightarrow C$ ).

```
class Animal:
  proc init(self, name):
    self.name = name
  proc speak(self):
    print self.name + " speaks"

class Dog(Animal):
  proc init(self, name, breed):
    super.init(name)
    self.breed = breed
  proc speak(self):
    super.speak()
    print self.name + " barks"

let d = Dog("Rex", "Labrador")
d.speak()
# Rex speaks
# Rex barks
```

The `->` arrow operator can also be used with `super`: `super->init(args)`.

### Arrow Operator (->):

The `->` operator is a syntactic alias for `.` (dot), providing systems-language style pointer/member access. It works identically for field reads, field writes, and method calls.

```
let p = Point(3, 7)
print p->x           # same as p.x
p->x = 10             # same as p.x = 10
print p->to_string()  # same as p.to_string()
```

`->` is interchangeable with `.` everywhere: attribute access, method calls, and `super` calls (`super->init(self, args)`).

## 4.5 Exception Handling

### Try-Catch-Finally:

```
proc safe_divide(a, b):
  if b == 0:
    raise "Zero denominator"
  return a / b

try:
  let result = safe_divide(10, 0)
catch error:
  print "Error: " + error
finally:
  print "Done"
```

```
# Output:
# Error: Zero denominator
# Done
```

**Finally Block Semantics:** The finally block always executes, and its control flow takes precedence over try/catch. If finally contains `return`, `break`, `continue`, or `raise`, that overrides the try/catch result. If finally executes normally, the try/catch result is preserved. This matches Python/Java behavior.

**Raise:** You can raise any value. Strings become exception messages directly. Numbers, booleans, and nil are converted to their string representation. Non-string/non-exception values become “Unknown error”.

```
raise "file not found"      # Exception with message "file not found"
raise 404                   # Exception with message "404"
raise nil                   # Exception with message "nil"
```

## 4.6 Generators

### Simple Generator:

```
proc fibonacci(n):
  let a = 0
  let b = 1
  let count = 0
  while count < n:
    yield a
    let temp = a + b
    a = b
    b = temp
    count = count + 1

let gen = fibonacci(5)
print next(gen)      # 0
print next(gen)      # 1
print next(gen)      # 1
print next(gen)      # 2
print next(gen)      # 3
print next(gen)      # nil
```

## 4.7 Modules

`math.sage:`

```

let PI = 3.14159

proc square(x):
    return x * x

proc cube(x):
    return x * x * x

proc factorial(n):
    if n <= 1:
        return 1
    return n * factorial(n - 1)

main.sage:

import math as m

print m.PI                # 3.14159
print m.square(5)         # 25
print m.cube(3)           # 27
print m.factorial(5)      # 120

# Or:
from math import PI, square, cube
print PI
print square(5)
print cube(3)

```

## 4.8 Defer Statements

Defer schedules cleanup code to run when the enclosing block exits, regardless of whether exit is via normal flow, **return**, **break**, **continue**, or exception. Multiple defers execute in LIFO (last-in, first-out) order.

```

proc process_file(path):
    let handle = open(path)
    defer:
        close(handle)
        print("file closed")
    # ... work with handle ...
    # close() runs automatically when block exits

proc multi_defer():
    defer:
        print("third")
    defer:
        print("second")
    defer:
        print("first")
    print("working")
    # Output: working, first, second, third

```

Defer works with all exit paths: - Normal block completion - **return** from a function - **break/continue** from a loop - Exceptions (defers run before exception propagates)

Implementation: the interpreter collects deferred statements in a 1024-slot stack per block. On block exit, they execute in reverse order.

## 4.9 Pattern Matching (match/case/default)

Match evaluates a value and compares it against case patterns using equality. The first matching case body executes. An optional **default** clause runs if no case matches.

```
let cmd = "hello"
match cmd:
  case "quit":
    print("quitting")
  case "hello":
    print("greeting")
  case "help":
    print("showing help")
  default:
    print("unknown command")
```

# Output: greeting

Match works with numbers, strings, booleans, and nil:

```
proc classify(n):
  match n:
    case 1:
      return "small"
    case 2:
      return "medium"
    default:
      return "big"
```

```
print(classify(1))    # small
print(classify(100))  # big
```

Cases are checked in order using value equality (==). Match is supported in the interpreter, C compiler (emits if/else-if chain), LLVM backend (emits branch instructions), and native codegen.

---

## Part 5: Implementation Details and Runtime Behavior

### 5.1 Memory Layout and Object Representation

**Stack vs. Heap:** - **Stack-allocated** (cheap, temporary): **Value** struct for numbers, bools, nil, function pointers. - **Heap-allocated** (managed by GC): Arrays, dicts, tuples, strings, classes, instances, exceptions, generators.

**Value Encoding** (union-based):

```
Value v;
v.type = VAL_NUMBER;
v.as.number = 42.5;           // Efficient packing; only one field used per type
```

*// Or:*

```
v.type = VAL_ARRAY;
v.as.array = malloc(...);    // Pointer to ArrayValue struct
```

**Implications:** - Numbers and bools are always “cheap” to copy (no allocation). - Strings are immutable; reassignment creates new allocation (handled by parser/interpreter). - Arrays/dicts are mutable and passed by reference (modifications visible across scopes).

## 5.2 Call Stack and Environments

**On Function Call:** 1. **Create child environment:** `new_env = env_create(current_env)`. 2. **Bind parameters:** `env_define(new_env, param_name, param_value)`. 3. **Execute body:** `interpret(body, new_env)`. 4. **Return:** Pop environment, return to caller.

**Nested Example:**

```
Global: { x: 10 }
  `~ call add(3, 4)
    `~ Local: { a: 3, b: 4 } (parent: Global)
      `~ return a + b (7)
```

## 5.3 Exception Propagation

**Raise Mechanism:** - `raise "message"` or `raise exception_obj` → Sets `is_throwing = 1` in `ExecResult`. - `raise` converts non-string values: numbers become their string representation, booleans become “true”/“false”, nil becomes “nil”. - All statements/expressions check `is_throwing` and propagate up. - First matching `catch` clause executes; if none, exception exits function. - Sage uses generic catch (no type-based matching). All catches handle all exceptions.

**Finally Guarantee:** - `finally` block always runs, even if exception, return, break, or continue occurred in try/catch. - If `finally` raises an exception or returns, that overrides the try/catch result (matches Python/Java semantics). - If `finally` completes normally, the original try/catch result is preserved.

**Memory Safety:** - Exception messages are allocated outside the GC heap but tracked via `gc_track_external_allocation()`. - GC properly frees exception messages with `gc_track_external_free()` during collection. - `VAL_EXCEPTION` objects are marked during GC mark phase to prevent premature collection.

## 5.4 Generator State Management

**Creation:** - `proc gen_func(): yield 1; yield 2` parses as normal proc. - **Call** returns `VAL_GENERATOR` with: - **body:** Pointer to function body (`Stmt*`). - **closure:** Captured environment at call time. - **is\_started:** 0 initially. - **is\_exhausted:** 0 initially. - **current\_stmt:** NULL (set on first `next()`).

**Resumption:** - **First next():** Initialize `gen_env` as child of closure, run body until first yield. - **Subsequent next():** Resume from `current_stmt` (set by prior yield), run until next yield or end. - **Yield execution:** Save `current_stmt = stmt->next`, return yielded value with `is_yielding = 1`.

**Example Trace:**

```
gen = my_gen()           # Create generator, is_started=0
next(gen)                 # Initialize gen_env, run to first yield
                           # Yield 10; current_stmt = stmt after yield
next(gen)                 # Resume from current_stmt, run to next yield
                           # Yield 20; current_stmt = stmt after yield
next(gen)                 # Resume, reach end, is_exhausted=1
                           # Return nil
```

## 5.5 Operator Precedence and Associativity

**Precedence** (lowest to highest): 1. Assignment (`=`) 2. Logical OR (`or`) 3. Logical AND (`and`) 4. Bitwise OR (`|`) 5. Bitwise XOR (`^`) 6. Bitwise AND (`&`) 7. Equality (`==`, `!=`) 8. Comparison (`<`, `>`, `<=`, `>=`) 9. Shift (`<<`, `>>`) 10. Addition/Subtraction (`+`, `-`) 11. Multiplication/Division (`*`, `/`, `%`) 12. Unary (`-`, `not`, `~`) 13. Postfix (indexing, slicing, property access) 14. Primary (literals, variables, parens, function calls)

**Associativity:** - **Left:** Most binary operators (`+`, `-`, `*`, `/`, `==`, `!=`, `<`, `>`, etc.). - **Right:** Assignment (`=`), unary (`-`).

## 5.6 Type Coercion and Truthiness

**Truthiness** (for if/while conditions): - **Falsy**: nil, false. - **Truthy**: Everything else (0 is truthy, empty arrays are truthy, etc.).

**Type Coercion**: - **Addition**: If both are numbers, add; if either is string, concatenate; else nil. - **Comparison**: Numbers compared numerically; strings compared lexically; different types are unequal. - **String conversion**: `str()` native function converts any value to string.

---

## Part 6: Compilation and Execution

### 6.1 Building the Interpreter

The repository exposes four build paths from source:

- Desktop build from C sources, which produces `sage` and `sage-lsp`
- Self-hosted bootstrap mode, which still builds `sage` first and then uses it to execute `src/sage/sage.sage`
- Pico/RP2040 builds through CMake and the Pico SDK
- SageMake: unified build system with auto-detection of platform, GPU (cuBLAS), NPU (NNAPI/SNPE/ONE), SIMD (NEON/RVV), and compiler backends

**Desktop Build (Make):**

```
make clean && make -j$(nproc)
./sage examples/hello.sage
```

**ML Trainer Build:**

```
make train-c          # Builds standalone C training binary (no Sage runtime); uses src/c/ml_backend.c dire
```

**Desktop Build (CMake):**

```
cmake -B build
cmake --build build
```

**SageMake Build:**

```
./sagemake info          # Show detected environment (GPU, NPU, SIMD, compiler)
./sagemake build          # Build sage interpreter
./sagemake chatbot --llvm # Compile chatbot via LLVM
./sagemake chatbot --c    # Compile via C backend
./sagemake chatbot --native # Compile via native asm
./sagemake train 200000 0.001 # Build trainer + train
./sagemake all            # Build everything
./sagemake --minimal build # Core only (no optional deps)
```

`./sagemake info` auto-detects GPU (cuBLAS), NPU (NNAPI/SNPE/ONE), SIMD (NEON/RVV), and selects optimal compilation flags.

**Self-Hosted Build / Bootstrap:**

```
make sage-boot FILE=examples/hello.sage
make test-selfhost
```

*# Or via CMake:*

```
cmake -B build_sage -DBUILD_SAGE=ON
cmake --build build_sage
cmake --build build_sage --target test_selfhost
```

**Pico Build:**

```
cmake -B build_pico -DBUILD_PICO=ON -DPICO_BOARD=pico
cmake --build build_pico
```

Desktop builds require `libcurl` and `OpenSSL` development headers/libraries in addition to a C compiler, `make`, and/or `cmake`.

### 3 Build Parameter Reference Make Variables:

| Variable   | Default                                                               | Effect                                                                              |
|------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| CC         | gcc                                                                   | C compiler used by <code>make</code>                                                |
| CFLAGS     | -std=c11 -Wall -Wextra<br>-Wpedantic -O2<br>-D_POSIX_C_SOURCE=200809L | Base compile flags for desktop builds                                               |
| LDFLAGS    | -lm -lpthread -ldl -lcurl<br>-lssl -lcrypto                           | Desktop link flags; reduced to <code>-lm</code> when <code>PICO_BUILD</code> is set |
| DEBUG      | 0                                                                     | <code>DEBUG=1</code> adds <code>-g -O0 -DDEBUG</code>                               |
| PREFIX     | /usr/local                                                            | Install prefix used by <code>make install</code>                                    |
| FILE       | unset                                                                 | Required by <code>make sage-boot</code> <code>FILE=&lt;path&gt;</code>              |
| PICO_BUILD | unset                                                                 | Internal Make switch that changes link flags for non-desktop builds                 |

### CMake Cache Variables / Environment Inputs:

| Parameter            | Default           | Effect                                                                                                  |
|----------------------|-------------------|---------------------------------------------------------------------------------------------------------|
| BUILD_PICO           | OFF               | Enables Pico/RP2040 output and imports <code>pico_sdk_import.cmake</code> before <code>project()</code> |
| BUILD_SAGE           | OFF               | Enables bootstrap/self-hosted targets such as <code>sage_boot</code> and <code>test_selfhost</code>     |
| ENABLE_DEBUG         | OFF               | Adds <code>-g -O0 -DDEBUG</code>                                                                        |
| ENABLE_TESTS         | OFF               | Builds optional C test executables                                                                      |
| CMAKE_BUILD_TYPE     | generator default | Standard CMake build-type selector                                                                      |
| CMAKE_C_COMPILER     | toolchain default | Chooses the C compiler reported in the configuration summary                                            |
| CMAKE_INSTALL_PREFIX | CMake default     | Install destination for <code>cmake --install</code>                                                    |
| PICO_SDK_PATH        | unset             | Required for Pico builds unless already exported in the environment                                     |
| PICO_BOARD           | pico              | Pico board name for SDK builds                                                                          |

| Parameter | Default | Effect                                                          |
|-----------|---------|-----------------------------------------------------------------|
| SAGE_FILE | unset   | Input path consumed by the <code>sage_boot</code> custom target |

## 6.2 Main Entry Point (main.c)

`src/c/main.c` initializes the garbage collector, registers raw `argv` for the `sys` module, initializes the module cache, and then dispatches one of the top-level modes below.

## 3 sage CLI Parameter Reference

| Command                                       | Meaning                              | Notes                                                                |
|-----------------------------------------------|--------------------------------------|----------------------------------------------------------------------|
| <code>sage</code>                             | Start the interactive REPL           | Same as <code>sage --repl</code>                                     |
| <code>sage --repl</code>                      | Start the interactive REPL           | Supports multi-line blocks and recovery after REPL errors            |
| <code>sage --help</code>                      | Print usage text                     | Covers compiler, tooling, and REPL entry points                      |
| <code>sage -c "source"</code>                 | Execute a source string              | No file is loaded                                                    |
| <code>sage &lt;file.sage&gt; [arg ...]</code> | Execute a Sage file                  | Additional CLI arguments are visible through <code>sys.args()</code> |
| <code>sage &lt;file.sgvml&gt;</code>          | Execute an SGVM binary               | Runs via integrated MetalVM engine                                   |
| <code>sage --lsp</code>                       | Start the LSP server on stdin/stdout | <code>sage-lsp</code> is the standalone wrapper binary               |
| <code>sage fmt &lt;file&gt;</code>            | Format a file in place               | Prints <b>Formatted:</b> <code>&lt;file&gt;</code> on success        |
| <code>sage fmt --check &lt;file&gt;</code>    | Check formatting without rewriting   | Exit code 1 when changes are needed                                  |
| <code>sage lint &lt;file&gt;</code>           | Run the static linter                | Exit code 1 when issues are found                                    |

| Compiler Command                                    | Default Output                           | Options                                                                                                                                                         |
|-----------------------------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>sage --emit-vm &lt;input.sage&gt;</code>      | <code>&lt;input&gt;.svm</code>           | <code>-o &lt;path&gt;</code> , <code>-O0</code> , <code>-O1</code> , <code>-O2</code> , <code>-O3</code> , <code>-g</code>                                      |
| <code>sage --sgvm &lt;input.sage&gt;</code>         | <code>&lt;input&gt;.sgvm</code>          | <code>-o &lt;path&gt;</code> , <code>-O0</code> , <code>-O1</code> , <code>-O2</code> , <code>-O3</code> , <code>-g</code>                                      |
| <code>sage --compile &lt;input.sage&gt;</code>      | <code>&lt;input-without-.sage&gt;</code> | <code>-o &lt;path&gt;</code> , <code>--cc &lt;compiler&gt;</code> , <code>-O0</code> , <code>-O1</code> , <code>-O2</code> , <code>-O3</code> , <code>-g</code> |
| <code>sage --emit-llvm &lt;input.sage&gt;</code>    | <code>&lt;input&gt;.ll</code>            | <code>-o &lt;path&gt;</code> , <code>-O0</code> , <code>-O1</code> , <code>-O2</code> , <code>-O3</code> , <code>-g</code>                                      |
| <code>sage --compile-llvm &lt;input.sage&gt;</code> | <code>&lt;input-without-.sage&gt;</code> | <code>-o &lt;path&gt;</code> , <code>-O0</code> , <code>-O1</code> , <code>-O2</code> , <code>-O3</code> , <code>-g</code>                                      |

| Compiler Command                                                    | Default Output                                                                    | Options                                                                                             |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| sage --emit-asm <input.sage>                                        | <input>.s                                                                         | -o <path>,<br>--target<br><arch[-profile]>,<br>-00, -01, -02, -03,<br>-g                            |
| sage --compile-native <input.sage>                                  | hosted: <input-without-.sage>;<br>non-hosted profiles:<br><input-without-.sage>.o | -o <path>,<br>--target<br><arch[-profile]>,<br>-00, -01, -02, -03,<br>-g                            |
| sage --emit-pico-c <input.sage><br>sage --compile-pico <input.sage> | <input>.pico.c<br>.tmp/<program-name> plus<br><program-name>.uf2                  | -o <path><br>-o <dir>,<br>--board <name>,<br>--name<br><program>, --sdk<br><path>, --chip<br><type> |
| --chip <type>                                                       | --compile-pico                                                                    | Chip type (rp2040,<br>rp2350-arm, or<br>rp2350-riscv);<br>defaults to rp2040                        |
| sage --compile-bare <input.sage>                                    | <input-without-.sage>.elf                                                         | -o <path>,<br>--target <arch>,<br>-00, -01, -02, -03,<br>-g                                         |
| sage --compile-uefi <input.sage>                                    | <input-without-.sage>.efi                                                         | -o <path>,<br>--target<br>x86_64\ aarch64,<br>-00, -01, -02, -03,<br>-g                             |
| sage --jit <input.sage>                                             | JIT executable (temporary)                                                        | -o <path> for<br>self-extracting<br>executable with<br>module bundling                              |

| Option                       | Applies To                                             | Meaning                                                                                                                                     |
|------------------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| -o <path>                    | All emit/compile commands, including<br>--compile-pico | Output file or build directory                                                                                                              |
| --cc <compiler>              | --compile                                              | Overrides the host C compiler;<br>defaults to cc                                                                                            |
| --target<br><arch[-profile]> | --emit-asm, --compile-native                           | Base targets: x86-64, x86_64,<br>aarch64, arm64, rv64,<br>riscv64, mips, mips32,<br>mips74k; profile suffixes:<br>-baremetal, -osdev, -uefi |
| -00 / -01 / -02 / -03        | C, LLVM, and native codegen commands                   | Selects the optimization pass<br>level                                                                                                      |
| -g                           | C, LLVM, asm, and native codegen<br>commands           | Enables debug information in<br>generated output                                                                                            |

| Option                              | Applies To                  | Meaning                                                                                                                   |
|-------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <code>--math-work=MODES</code>      | All runtime modes           | Comma-separated list of default math visualization formats: <code>grade</code> , <code>exec</code> , <code>bitwise</code> |
| <code>--board &lt;name&gt;</code>   | <code>--compile-pico</code> | Pico board name; defaults to <code>pico</code>                                                                            |
| <code>--name &lt;program&gt;</code> | <code>--compile-pico</code> | Overrides the generated program name derived from the input file                                                          |
| <code>--sdk &lt;path&gt;</code>     | <code>--compile-pico</code> | Pico SDK path; falls back to <code>PICO_SDK_PATH</code>                                                                   |

LLVM backend notes:

- `--compile-llvm` produces a fully native binary by emitting LLVM IR and invoking `clang` to compile and link it. The result is a standalone executable with no Sage runtime dependency.
- As a real-world data point, the SageLLM chatbot (`models/chatbots/sagellm_chatbot.sage`) compiles to a **124 KB standalone binary** via `--compile-llvm`, including the full inference loop and tokenizer.

Profile notes for native backend:

- `hosted` (no suffix): default behavior.
- `-baremetal` / `-osdev`: emits `sage_entry` and freestanding object-oriented output.
- `-uefi`: emits `efi_main`; current stage emits a freestanding object, with full PE/COFF image linking planned as follow-up work.

### 6.3 Program Execution Flow

1. **Read source** from file.
2. **Initialize lexer** with source string.
3. **Initialize parser** (advance to first token).
4. **Create global environment** and initialize `stdlib`.
5. **Parse declarations** (statements and function definitions).
6. **Execute each statement** using the selected runtime backend.
7. **Return** at EOF.

### 6.4 Runtime Backends

The C-hosted `sage` binary now supports several runtime selections:

| Mode                  | Command                                        | Current Behavior                                                            |
|-----------------------|------------------------------------------------|-----------------------------------------------------------------------------|
| <code>ast</code>      | <code>sage --runtime ast file.sage</code>      | Original tree-walking interpreter; highest maturity and easiest to debug    |
| <code>bytecode</code> | <code>sage --runtime bytecode file.sage</code> | Lowers each top-level statement to bytecode and executes it on the stack VM |
| <code>auto</code>     | <code>sage --runtime auto file.sage</code>     | Default; resolves to JIT on hosted, AST on bare-metal                       |
| <code>jit</code>      | <code>sage --runtime jit file.sage</code>      | Enables JIT profiling and type feedback                                     |
| <code>aot</code>      | <code>sage --runtime aot file.sage</code>      | Enables Ahead-of-Time type specialization                                   |

### Current VM architecture:

- The lexer and parser are unchanged; the VM reuses the same AST front-end.
- Each parsed top-level statement is compiled to a transient bytecode chunk, then executed immediately.
- The C-hosted toolchain can also emit a strict ahead-of-time VM artifact with `sage --emit-vm file.sage` and execute it later with `sage --run-vm file.svm`.
- The self-hosted CLI can emit the same artifact format with `sage sage.sage --emit-vm file.sage`, including compiled proc bodies and returns.
- `sage --runtime bytecode` remains **hybrid** today: unsupported top-level statements fall back to the AST interpreter through an explicit AST-bridge opcode instead of failing the whole program.
- `sage --emit-vm` is intentionally stricter: unsupported constructs fail compilation instead of silently bridging at runtime.
- Values, environments, modules, classes, instances, and the GC are shared between AST and bytecode execution.

### What runs natively in the VM today:

- Literals (`number`, `string`, `bool`, `nil`)
- Global variables (`let`, assignment, reads)
- Arithmetic, comparison, logical, and bitwise operators
- Arrays, tuples, dicts, indexing, slicing, and property access
- `print`, expression statements, `if`, `while`, and array `for`
- `break` and `continue` inside while/for loops (compiled natively with loop context stack)
- Ahead-of-time proc definitions, proc calls, nested proc calls, explicit `return`, and implicit `nil` returns
- Calls to native functions, Sage functions, classes, and instance methods
- 30 GPU hot-path opcodes for frame-loop performance (`poll_events`, `key_pressed`, `cmd_draw`, etc.)

### What still bridges or stays unsupported:

- In hybrid `--runtime bytecode` mode: class definitions, module imports, exception handling (`try/catch/raise`), `defer`, `match`, `yield`, and `async` procs fall back to the AST interpreter via `BC_OP_EXEC_AST_STMT`. Opcodes are defined for future native support (`BC_OP_CLASS`, `BC_OP_IMPORT`, `BC_OP_SETUP_TRY`, `BC_OP_RAISE`, etc.).
- In strict `--emit-vm` mode: these constructs fail compilation instead of bridging.
- `EXPR_AWAIT` is not supported in either mode.

**Security:** The VM validates all constant pool accesses (`VM_CHECK_CONST`) and AST statement indices (`VM_CHECK_AST`) to prevent buffer overflow from malformed bytecode. Stack depth is bounded at 1024 entries. All memory allocation uses OOM-safe wrappers.

The practical result is that `bytecode` mode is already useful for long-running scripts and engine-style workloads, `--emit-vm` is now a real ahead-of-time path for a meaningful strict subset of Sage, and `ast` mode remains the reference path for maximum behavioral confidence.

---

## Part 7: Advanced Topics and Internals

### 7.1 Memory Leaks and GC Limitations

**Known Limitations:**

- **String interning:** Strings are not interned; identical strings create separate allocations.
- **Non-GC allocations:** AST nodes, token buffers, and other manual allocations outside the `Value` heap still need explicit ownership and cleanup discipline.
- **Manual `malloc`:** Some internal allocations (e.g., AST nodes, Token data) use `malloc` without GC tracking. Only heap-allocated `Values` are GC'd.

#### Cycle collection:

- The GC is now a tracing mark-and-sweep collector that can reclaim **circular references** among GC-managed Sage values such as arrays, dicts, tuples, classes, instances, functions, and generators.

- Cycles are only collectable when every object in the cycle is represented in the managed **Value** graph. Raw pointers, external allocations, and ad hoc native-side ownership still sit outside that guarantee.

#### LLVM Backend Loop Variable Limitation:

- **For-loop iteration variables cannot be modified to simulate a break in LLVM-compiled binaries.** Code like `j = len(arr)` inside a for loop has no effect on the loop control variable in `--compile-llvm` output. Always use `break` to exit loops early. This works correctly in the interpreter and all other compiler backends (C, native).

**Practical Tips:** - Call `gc_collect()` periodically in long-running loops to prevent heap explosion. - Avoid creating large temporary structures in loops. - Reuse arrays/dicts by modifying in-place rather than creating new ones.

## 7.2 Performance Characteristics

**Interpreter Overhead:** - **AST runtime:** Walks the parsed tree directly. It is still the simplest and most mature execution path. - **Bytecode VM:** Adds a statement-at-a-time lowering step, then executes stack bytecode. On mixed arithmetic/loop workloads it is already modestly faster than AST mode. - **Threaded Dispatch:** The VM uses computed gotos (threaded interpretation) for high-performance instruction dispatch on supported compilers, significantly reducing loop overhead. - **Register Spilling:** The VM stack pointer and instruction pointer are kept in CPU registers where possible, minimizing memory traffic during the execution loop. - **Nested scopes:** Linear search up parent chain;  $O(\text{depth})$  for variable lookup. - **GC pauses:** Mark-and-sweep still pauses execution during collection, but the trigger is now dynamic and based on both object count and managed bytes.

**Optimization Opportunities:** - Bytecode functions and method bodies, so fewer calls bridge back to AST. - Local-slot or register-based VM frames instead of env-backed global lookups for more code. - Variable lookup caching (memoization). - Generational GC (mark only young objects frequently). - JIT compilation for hot paths.

## 3 Current Recipe Benchmark

The repository now includes a five-recipe benchmark:

```
python3 scripts/benchmark_recipes.py --runs 5 --warmups 1
```

It measures:

- **sage-interpreted-c-ast:** the C-hosted tree-walking interpreter
- **sage-interpreted-vm:** the C-hosted bytecode VM runtime
- **sage-compiled-c:** the native C backend compiled to a host executable
- **sage-compiled-vm:** the C-hosted ahead-of-time VM artifact (`--emit-vm + --run-vm`)
- **sage-compiled-sage:** the self-hosted Sage compiler path that emits C and then compiles it with the host toolchain

Workload source: `benchmarks/runtime_compare.sage`

Generated chart assets:

- `assets/charts/benchmark-recipes-total.svg`
- `assets/charts/benchmark-recipes-run.svg`

Current caveats:

- **sage-compiled-vm** now passes the default benchmark workload and is charted as a real execution lane, but it is still a strict subset backend: unsupported constructs fail during `--emit-vm` instead of falling back at runtime.
- **sage-compiled-sage** is still experimental and currently fails checksum validation on the default benchmark workload, so it is called out in the chart footer instead of being charted as a valid timing result.

Interpretation:

- `sage-interpreted-vm` is the direct runtime-backend comparison against `sage-interpreted-c-ast`.
- `sage-compiled-c` has the lowest execution-only runtime on the default workload, but its total wall time includes code generation and host compilation.
- The total-time chart answers “time to result”; the execution-only chart answers “steady-state runtime after the binary already exists.”

### 3 Sage vs Python 3 Benchmarks

A separate benchmark suite compares all Sage execution paths against CPython 3.x:

```
make benchmark-python      # Console table output
make benchmark-python-md   # Markdown table output
```

10 paired workloads (benchmarks/01\_fibonacci.sage + .py through benchmarks/10\_primes\_sieve):

| Benchmark          | What it tests                         |
|--------------------|---------------------------------------|
| fibonacci          | Recursive function call overhead      |
| loop_sum           | Raw loop + arithmetic throughput      |
| string_concat      | String allocation + GC pressure       |
| array_ops          | Dynamic array growth + iteration      |
| dict_ops           | Hash table insert + lookup            |
| class_method       | OOP method dispatch                   |
| nested_loops       | Control flow with break/continue      |
| exception_handling | Try/catch overhead                    |
| recursion_closures | Closure creation + higher-order calls |
| primes_sieve       | Array access + conditional logic      |

Five recipes tested: Python 3, Sage AST, Sage VM, Sage C (compiled), Sage LLVM (compiled).

Typical performance characteristics: - **LLVM compiled**: 2-8x faster than CPython (fibonacci 3.9x, loop sum 7.7x) - **C compiled**: 2-4x faster than CPython on most workloads - **AST interpreter**: On par with CPython (faster on string/array ops, slower on deep recursion) - **Bytecode VM**: Similar to AST, with measurable speedups on tight loops

### 7.3 Extending SageLang with Native Functions

Adding a Native Function:

```
// In interpreter.c, add:
static Value my_function(int argCount, Value* args) {
    if (argCount != 2) return val_nil();
    if (!IS_NUMBER(args[0]) || !IS_NUMBER(args[1])) return val_nil();
    double result = AS_NUMBER(args[0]) + AS_NUMBER(args[1]) * 2;
    return val_number(result);
}
```

```
// In init_stdlib(), add:
env_define(env, "my_func", 7, val_native(my_function));
```

Usage from SageLang:

```
print my_func(3, 4) # 3 + 4*2 = 11
```

## 7.4 Debugging and Introspection

**Print Value:**

```
void print_value(Value v); // In value.c
```

**GC Stats:**

```
let stats = gc_stats()
print stats # {"bytes_allocated": ..., "num_objects": ..., ...}
```

**Tracing Execution** (manual): - Add `printf()` statements to `interpreter.c` before/after statement/expression evaluation. - Mark garbage objects: Modify `gc_sweep()` to not free unmarked objects, inspect them.

## 7.5 Interpreter Safety Limits

SageLang enforces several compile-time and runtime limits to prevent crashes from malicious or malformed input:

| Limit                 | Constant                         | Value   | Location                   | Behavior on Violation |
|-----------------------|----------------------------------|---------|----------------------------|-----------------------|
| Recursion depth       | <code>MAX_RECURSION_DEPTH</code> | 100000  | <code>interpreter.c</code> | Catchable exception   |
| Parser nesting        | <code>MAX_PARSER_DEPTH</code>    | 100000  | <code>parser.c</code>      | Parse error           |
| Loop iterations       | <code>MAX_LOOP_ITERATIONS</code> | 500,000 | <code>interpreter.c</code> | Catchable exception   |
| String literal length | <code>MAX_STRING_LENGTH</code>   | 1096    | <code>lexer.c</code>       | Parse error           |
| Function arguments    | Stack array                      | 255     | <code>interpreter.c</code> | Runtime error         |

**Null function guards:** The interpreter checks for null pointers in both `VAL_FUNCTION` and `VAL_NATIVE` call paths before dispatch. A null callee produces a runtime error and returns `nil`.

**Type-safe accessor macros** (`value.h`):

| Macro                          | Returns on type mismatch       |
|--------------------------------|--------------------------------|
| <code>SAGE_AS_STRING(v)</code> | <code>""</code> (empty string) |
| <code>SAGE_AS_NUMBER(v)</code> | <code>0.0</code>               |
| <code>SAGE_AS_BOOL(v)</code>   | <code>0</code> (false)         |

These are intended for native function implementations that want defensive coercion. The standard `IS_*/AS_*` pattern (check type, then access) remains the recommended approach for most code:

```
static Value my_native(int argCount, Value* args) {
    if (argCount < 1 || !IS_STRING(args[0])) return val_nil();
    const char* s = AS_STRING(args[0]); // safe - guarded by IS_STRING above
    // ...
}
```

## Part 8: Future Directions and Design Notes

### 8.1 Planned Features (Incomplete Implementations)

**Pattern Matching** (`match/case`): Fully implemented across all backends (see Section 4.9).

**Defer Statements**: Fully implemented with LIFO ordering and scope-exit semantics (see Section 4.8).

**Multiple Inheritance / Traits**: - Currently single-parent inheritance only.

**Operator Overloading**: - No support for custom `__add__()`, etc.

**Type Annotations**: - Fully supported for variables (`let x: Int = 42`) and procedures (`proc add(a: Int, b: Int) -> Int:`). - Dotted names are supported for qualified types (e.g., `let x: vfs.VFS`). - Type checking is enforced during a static analysis pass (`pass_typecheck` in `typecheck.c`) prior to compilation or interpretation.

### 8.2 Design Decisions and Rationale

**Why Indentation-Based Syntax?** - Python familiarity for embedded systems developers. - Reduces bracket clutter in embedded/IoT code. - Lexer complexity worth the syntactic clarity.

**Why Mark-and-Sweep Over Reference Counting?** - Simpler to implement; no cycle detection overhead. - Acceptable for embedded contexts where pause is tolerable. - Future: Generational GC for better pause times.

**Why Tree-Walking Interpreter?** - Rapid prototyping and debugging. - Minimal complexity; easy to extend. - Suitable for small programs; not for performance-critical workloads.

**Why Classes Over Functions-as-Objects?** - Explicit OOP model matches C++ style (familiar to embedded developers). - Clearer semantics for `self` and method dispatch.

---

## Part 9: Bundled Library Modules (`lib/`)

SageLang ships with a set of **pure-Sage library modules** in the `lib/` directory. These are importable via `import` or `from ... import` when running from the project root (the module resolver searches `./lib/`).

**Note:** `lib/math.sage` and native `math` share a name — the native module takes precedence. Use `import math` for native math functions (`sin`, `cos`, `sqrt`, etc.) and `from math import ...` for native `math`. The `lib/math.sage` helpers (`factorial`, `gcd`, etc.) are available if no native module shadows them.

### 9.1 Arrays Module (`lib/arrays.sage`)

Functional array utilities:

```
from arrays import map, filter, reduce, reverse, unique, flatten, zip, chunk
```

```
proc double(x):  
    return x * 2
```

```
print map([1, 2, 3], double)      # [2, 4, 6]  
print filter([1, 2, 3, 4], is_even) # [2, 4]  
print reverse([1, 2, 3])          # [3, 2, 1]  
print unique([1, 2, 2, 3])         # [1, 2, 3]  
print flatten([[1, 2], [3, 4]])    # [1, 2, 3, 4]  
print zip([1, 2], ["a", "b"])       # [(1, a), (2, b)]  
print chunk([1, 2, 3, 4, 5], 2)     # [[1, 2], [3, 4], [5]]
```

| Function                               | Description                      |
|----------------------------------------|----------------------------------|
| <code>copy(arr)</code>                 | Shallow copy                     |
| <code>append_all(target, extra)</code> | Extend target in-place           |
| <code>concat(a, b)</code>              | Return new merged array          |
| <code>reverse(arr)</code>              | Return reversed copy             |
| <code>map(arr, fn)</code>              | Apply fn to each element         |
| <code>filter(arr, pred)</code>         | Keep elements where pred is true |
| <code>reduce(arr, init, fn)</code>     | Fold left with accumulator       |
| <code>contains(arr, val)</code>        | Check membership                 |
| <code>index_of(arr, val)</code>        | Find index (-1 if missing)       |
| <code>find(arr, pred)</code>           | First element matching predicate |
| <code>unique(arr)</code>               | Remove duplicates                |
| <code>flatten(nested)</code>           | Flatten one level of nesting     |
| <code>take(arr, n)</code>              | First n elements                 |
| <code>drop(arr, n)</code>              | Skip first n elements            |
| <code>zip(a, b)</code>                 | Pair elements as tuples          |
| <code>chunk(arr, size)</code>          | Split into chunks                |

## 9.2 Strings Module (`lib/strings.sage`)

String manipulation utilities:

```
from strings import words, compact, pad_left, dash_case, from_bin

print words(" hello world ") # [hello, world]
print compact(" hello world ") # hello world
print pad_left("42", 5, "0") # 00042
print dash_case("hello world") # hello-world
print from_bin("0b1010") # 10
```

| Function                                 | Description                          |
|------------------------------------------|--------------------------------------|
| <code>words(text)</code>                 | Split on whitespace, remove empties  |
| <code>compact(text)</code>               | Collapse whitespace to single spaces |
| <code>contains(text, sub)</code>         | Substring check                      |
| <code>count_substring(text, sub)</code>  | Count occurrences                    |
| <code>repeat(text, n)</code>             | Repeat string n times                |
| <code>pad_left(text, width, pad)</code>  | Left-pad to width                    |
| <code>pad_right(text, width, pad)</code> | Right-pad to width                   |
| <code>surround(text, left, right)</code> | Wrap with delimiters                 |
| <code>csv(values)</code>                 | Join with commas                     |
| <code>dash_case(text)</code>             | Convert to dash-case                 |
| <code>snake_case(text)</code>            | Convert to snake_case                |
| <code>endswith(text, suffix)</code>      | Check suffix                         |
| <code>from_bin(bits)</code>              | Binary string to number              |

## 9.3 Dicts Module (`lib/dicts.sage`)

Dictionary query and manipulation helpers:

```
from dicts import has, size, get_or, has_all, entries, remove_keys

let d = {}
d["x"] = 10
```

```
d["y"] = 20
```

```
print has(d, "x")           # true
print size(d)               # 2
print get_or(d, "z", "default") # default
print has_all(d, ["x", "y"]) # true
```

| Function                         | Description                  |
|----------------------------------|------------------------------|
| keys(d)                          | Wrapper for dict_keys        |
| values(d)                        | Wrapper for dict_values      |
| size(d)                          | Number of keys               |
| has(d, key)                      | Key existence check          |
| get_or(d, key, fallback)         | Get with default             |
| entries(d)                       | Array of (key, value) tuples |
| has_all(d, keys)                 | All keys present?            |
| has_any(d, keys)                 | Any key present?             |
| select_values(d, keys, fallback) | Get multiple values          |
| remove_keys(d, keys)             | Delete keys in-place         |
| count_missing(d, keys)           | Count absent keys            |

## 9.4 Iter Module (lib/iter.sage)

Reusable generator functions:

```
from iter import count, range_step, cycle, enumerate_array, take
```

```
let evens = range_step(0, 20, 2)
print take(evens, 5) # [0, 2, 4, 6, 8]
```

```
let c = cycle([1, 2, 3])
print take(c, 7) # [1, 2, 3, 1, 2, 3, 1]
```

```
let e = enumerate_array(["a", "b", "c"])
print next(e) # (0, a)
print next(e) # (1, b)
```

| Function                     | Description                     |
|------------------------------|---------------------------------|
| count(start, step)           | Infinite counter generator      |
| range_step(start, end, step) | Range with custom step          |
| repeat(value, count)         | Yield value n times             |
| repeat_forever(value)        | Infinite repeat                 |
| enumerate_array(arr)         | Yield (index, value) tuples     |
| cycle(arr)                   | Infinite cycle through array    |
| take(gen, n)                 | Collect n values from generator |
| nth(gen, index)              | Get nth value from generator    |

## 9.5 Stats Module (lib/stats.sage)

Statistical functions:

```
from stats import mean, variance, stddev, cumulative, normalize
```

```
print mean([1, 2, 3, 4, 5]) # 3
```

```

print variance([1, 2, 3, 4, 5]) # 2
print cumulative([1, 2, 3, 4, 5]) # [1, 3, 6, 10, 15]
print normalize([1, 2, 3, 4, 5]) # [0, 0.25, 0.5, 0.75, 1]

```

| Function                        | Description         |
|---------------------------------|---------------------|
| <code>sum(values)</code>        | Sum of array        |
| <code>product(values)</code>    | Product of array    |
| <code>min_value(values)</code>  | Minimum element     |
| <code>max_value(values)</code>  | Maximum element     |
| <code>mean(values)</code>       | Arithmetic mean     |
| <code>range_span(values)</code> | max - min           |
| <code>cumulative(values)</code> | Running totals      |
| <code>variance(values)</code>   | Population variance |
| <code>stddev(values)</code>     | Standard deviation  |
| <code>normalize(values)</code>  | Scale to [0, 1]     |

## 9.6 Utils Module (`lib/Utils.sage`)

General-purpose helpers:

```
from utils import default_if_nil, choose, between, head, last, repeat_value
```

```

print default_if_nil(nil, 42) # 42
print choose(true, "yes", "no") # yes
print between(5, 1, 10) # true
print head([1, 2, 3]) # 1
print last([1, 2, 3]) # 3
print repeat_value(0, 5) # [0, 0, 0, 0, 0]

```

| Function                                   | Description                       |
|--------------------------------------------|-----------------------------------|
| <code>identity(x)</code>                   | Return x unchanged                |
| <code>choose(cond, a, b)</code>            | Ternary selection                 |
| <code>default_if_nil(val, fallback)</code> | Nil coalescing                    |
| <code>is_even(n) / is_odd(n)</code>        | Parity check                      |
| <code>between(val, lo, hi)</code>          | Range check                       |
| <code>swap(a, b)</code>                    | Return (b, a) tuple               |
| <code>head(arr) / last(arr)</code>         | First/last element (nil if empty) |
| <code>repeat_value(val, n)</code>          | Array of n copies                 |
| <code>times(n, fn)</code>                  | Call fn(i) n times                |

## 9.7 Assert Module (`lib/assert.sage`)

Test assertion helpers (raise on failure):

```
from assert import assert_equal, assert_true, assert_close
```

```

assert_true(len([1, 2, 3]) == 3, "wrong length")
assert_equal(2 + 2, 4, "math is broken")
assert_close(3.14, 3.14159, 0.01, "not close enough")

```

| Function               | Description  |
|------------------------|--------------|
| <code>fail(msg)</code> | Always raise |

| Function                                              | Description                     |
|-------------------------------------------------------|---------------------------------|
| <code>assert_true(cond, msg)</code>                   | Raise if false                  |
| <code>assert_false(cond, msg)</code>                  | Raise if true                   |
| <code>assert_equal(actual, expected, msg)</code>      | Raise if not equal              |
| <code>assert_nil(val, msg)</code>                     | Raise if not nil                |
| <code>assert_not_nil(val, msg)</code>                 | Raise if nil                    |
| <code>assert_close(actual, expected, tol, msg)</code> | Raise if difference > tolerance |
| <code>assert_array_contains(arr, val, msg)</code>     | Raise if val not in arr         |

## 9.8 Math Module (`lib/math.sage`)

Pure-Sage math helpers (shadowed by native `math` module — use when native module is unavailable):

| Function                                       | Description                       |
|------------------------------------------------|-----------------------------------|
| <code>add, sub, mul, div</code>                | Basic arithmetic                  |
| <code>min, max, abs, sign</code>               | Comparison helpers                |
| <code>clamp(val, lo, hi)</code>                | Clamp to range                    |
| <code>square(x), cube(x)</code>                | Powers                            |
| <code>lerp(a, b, t)</code>                     | Linear interpolation              |
| <code>pow_int(base, exp)</code>                | Integer exponentiation            |
| <code>factorial(n)</code>                      | $n!$                              |
| <code>gcd(a, b), lcm(a, b)</code>              | GCD and LCM                       |
| <code>sum(arr), product(arr), mean(arr)</code> | Aggregates                        |
| <code>sqrt(n)</code>                           | Newton's method approximation     |
| <code>printm(expr, backend, formats)</code>    | Evaluate expression and show work |
| <code>distance(x1, y1, x2, y2)</code>          | Euclidean distance                |

### Arithmetic Visualization (`printm`):

The `math.printm()` function evaluates an arithmetic expression provided as a string and visualizes the step-by-step “work” for each operation.

```
import math
math.printm("123 + 456 * 2", backend="sage")
```

- **backend:**
  - “sage” (default): Pure SageLang implementation using vertical “grade-school” format.
  - “c”: Uses native C-level arithmetic bridges.
  - “asm”: Uses inline assembly (x86-64, aarch64, rv64, mips) and calls `printf` via FFI to show the instruction-level execution.
- **formats:** An array of strings to customize output.
  - “grade”: Vertical grade-school arithmetic visualization.
  - “exec”: Show the underlying execution mode (e.g., “[ASM] Using ADD instruction”).
  - “bitwise”: Reserved for bitwise operation visualization.

The default format can be configured globally using the `--math-work` CLI flag. | `normalize(val, lo, hi)` | Scale to `[0, 1]` |

## 9.9 GPU Libraries

SageLang ships with 18 GPU/rendering library modules in `lib/graphics/`. All are imported with the `graphics.` prefix (e.g., `import graphics.vulkan` binds as `vulkan`):

| Module                        | Import                                   | Lines | Purpose                                                                     |
|-------------------------------|------------------------------------------|-------|-----------------------------------------------------------------------------|
| <code>vulkan.sage</code>      | <code>import graphics.vulkan</code>      | ~425  | Ergonomic Vulkan builder API (string-based buffer/shader/pipeline creation) |
| <code>gpu.sage</code>         | <code>import graphics.gpu</code>         | ~150  | High-level compute helpers (one-shot dispatch, ping-pong buffers)           |
| <code>opengl.sage</code>      | <code>import graphics.opengl</code>      | ~400  | Drop-in OpenGL backend (same API as gpu module, OpenGL 4.5 init)            |
| <code>ui.sage</code>          | <code>import graphics.ui</code>          | ~400  | Immediate-mode GUI widgets (windows, buttons, sliders, menus, text inputs)  |
| <code>math3d.sage</code>      | <code>import graphics.math3d</code>      | ~250  | vec2/3/4, mat4, perspective/ortho projections, camera utilities             |
| <code>mesh.sage</code>        | <code>import graphics.mesh</code>        | ~300  | Procedural geometry (cube, plane, sphere), OBJ loading, GPU upload          |
| <code>renderer.sage</code>    | <code>import graphics.renderer</code>    | ~200  | Frame loop management (depth buffer, render pass, sync)                     |
| <code>material.sage</code>    | <code>import graphics.material</code>    | ~150  | Shader+texture+descriptor binding, material presets                         |
| <code>scene.sage</code>       | <code>import graphics.scene</code>       | ~200  | Node hierarchy, transforms, traversal, <code>find_by_name</code>            |
| <code>pbr.sage</code>         | <code>import graphics.pbr</code>         | ~300  | Cook-Torrance PBR materials, point/directional lights, IBL                  |
| <code>postprocess.sage</code> | <code>import graphics.postprocess</code> | ~250  | HDR targets, bloom chain, tone mapping (ACES/Reinhard)                      |
| <code>shadows.sage</code>     | <code>import graphics.shadows</code>     | ~200  | Shadow maps, depth-only passes, cascade shadow maps                         |
| <code>deferred.sage</code>    | <code>import graphics.deferred</code>    | ~300  | G-buffer (4 MRT), SSAO (32 samples), SSR (64-step raymarch)                 |
| <code>taa.sage</code>         | <code>import graphics.taa</code>         | ~150  | Temporal anti-aliasing (Halton jitter, history blend)                       |
| <code>gltf.sage</code>        | <code>import graphics.gltf</code>        | ~200  | glTF 2.0 JSON loading, mesh/material extraction                             |
| <code>asset_cache.sage</code> | <code>import graphics.asset_cache</code> | ~100  | Shader/texture/mesh caching and deduplication                               |
| <code>frame_graph.sage</code> | <code>import graphics.frame_graph</code> | ~150  | Pass dependency ordering (topological sort)                                 |
| <code>debug_ui.sage</code>    | <code>import graphics.debug_ui</code>    | ~150  | FPS tracking, custom debug values, toggle overlay                           |

## 9.10 OS Development Libraries

SageLang ships with 52 OS/bare-metal development modules across `lib/os/`, `lib/os/boot/`, `lib/os/kernel/`, and `lib/os/image/`:

| Module                    | Import                         | Purpose                                                                   |
|---------------------------|--------------------------------|---------------------------------------------------------------------------|
| <code>fat.sage</code>     | <code>import os.fat</code>     | FAT8/12/16/32 boot sector parser, cluster-to-LBA, FAT entry offsets       |
| <code>fat_dir.sage</code> | <code>import os.fat_dir</code> | FAT directory traversal, file reading, path resolution, cluster chains    |
| <code>elf.sage</code>     | <code>import os.elf</code>     | ELF32/64 header, program/section headers, string table lookup             |
| <code>mbr.sage</code>     | <code>import os.mbr</code>     | MBR partition table, CHS decode, bootable partition finder                |
| <code>gpt.sage</code>     | <code>import os.gpt</code>     | GPT header, GUID parsing, partition type identification                   |
| <code>pe.sage</code>      | <code>import os.pe</code>      | PE/COFF binary parser, DOS/COFF/optional headers, UEFI app detection      |
| <code>pci.sage</code>     | <code>import os.pci</code>     | PCI config space (Type 0/1), BAR decode, capability lists                 |
| <code>uefi.sage</code>    | <code>import os.uefi</code>    | EFI memory map, config tables, RSDP, ACPI SDT headers                     |
| <code>acpi.sage</code>    | <code>import os.acpi</code>    | MADT (APIC), FADT, HPET, MCFG parsers, processor enumeration              |
| <code>paging.sage</code>  | <code>import os.paging</code>  | x86-64 page tables, PTE flags, identity/higher-half mapping helpers       |
| <code>idt.sage</code>     | <code>import os.idt</code>     | x86-64 IDT gate construction, exception vectors, PIC remapping            |
| <code>serial.sage</code>  | <code>import os.serial</code>  | UART/COM port configuration, init sequences, debug output encoding        |
| <code>dtb.sage</code>     | <code>import os.dtb</code>     | Flattened Device Tree parser for ARM64/RISC-V (nodes, properties, search) |
| <code>alloc.sage</code>   | <code>import os.alloc</code>   | Bump, free-list, and bitmap page allocators for kernel memory             |
| <code>vfs.sage</code>     | <code>import os.vfs</code>     | Virtual filesystem layer with mount table, path utilities, memfs backend  |

| Module                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Import                                                                         | Purpose |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|---------|
| <pre> ext.sage"  import os.ext  ext2/3/4 superblock, inode table, directory entries, extent tree    btrfs.sage import os.btrfs  Btrfs superblock, chunk tree, root tree, subvolumes, checksums    f2fs.sage import os.f2fs  F2FS superblock, checkpoint, segment info, node/data addressing    boot/multiboot.sage import os.boot.multiboot  Multiboot2 header generation, tag building, boot info parsing    boot/gdt.sage import os.boot.gdt  x86_64 GDT descriptor construction, TSS entries, LGDT sequence    boot/start.sage import os.boot.start  x86_64 startup assembly generation (long mode entry, stack setup)    boot/linker.sage import os.boot.linker  Linker script generation for bare-metal ELF kernels    boot/bios.sage import os.boot.bios  BIOS interrupt interface (INT 0x10, 0x13, 0x15, 0x16) for legacy x86 boot    boot/e820.sage import os.boot.e820  BIOS memory map collection and normalization    boot/a20.sage import os.boot.a20  A20 gate enabling methods (BIOS, Fast A20, Keyboard Controller)    boot/cpuid.sage import os.boot.cpuid  CPU feature detection (Long mode, MSR, SSE, Vendor/Brand strings)    boot/real_mode.sage import os.boot.real_mode  Helpers for segment:offset addressing and real-mode stack setup    boot/prot_mode.sage import os.boot.prot_mode  Transition logic from real mode to 32-bit protected mode    boot/long_mode.sage import os.boot.long_mode  Transition logic from </pre> | <pre> Early VGA text-mode display, cursor management, and progress bars </pre> |         |

### 9.11 Networking Libraries

SageLang ships with 8 high-level networking modules in `lib/net/`, built on top of the native `socket`, `tcp`, `http`, and `ssl` modules:

| Module                      | Import                            | Purpose                                                                    |
|-----------------------------|-----------------------------------|----------------------------------------------------------------------------|
| <code>url.sage</code>       | <code>import net.url</code>       | URL parsing/building, percent-encoding/decoding, query strings             |
| <code>headers.sage</code>   | <code>import net.headers</code>   | HTTP header parsing/building, content-type inspection, constants           |
| <code>request.sage</code>   | <code>import net.request</code>   | HTTP request builder with fluent API, auth helpers, status utilities       |
| <code>server.sage</code>    | <code>import net.server</code>    | TCP/HTTP server framework with routing, request parsing, response builders |
| <code>websocket.sage</code> | <code>import net.websocket</code> | WebSocket frame building/parsing (RFC 6455), upgrade handshake             |
| <code>mime.sage</code>      | <code>import net.mime</code>      | MIME type lookup from file extensions (80+ types)                          |
| <code>dns.sage</code>       | <code>import net.dns</code>       | DNS wire-format message parsing/building, name compression                 |
| <code>ip.sage</code>        | <code>import net.ip</code>        | IPv4 parsing/validation, CIDR subnets, private/loopback/multicast checks   |

### 9.12 Cryptography Libraries

SageLang ships with 6 cryptography modules in `lib/crypto/`:

| Module                     | Import                              | Purpose                                                             |
|----------------------------|-------------------------------------|---------------------------------------------------------------------|
| <code>hash.sage</code>     | <code>import crypto.hash</code>     | SHA-256, SHA-1, CRC-32 hash functions with hex output               |
| <code>hmac.sage</code>     | <code>import crypto.hmac</code>     | HMAC (RFC 2104) with pluggable hash, constant-time comparison       |
| <code>encoding.sage</code> | <code>import crypto.encoding</code> | Base64 (standard + URL-safe), hex encoding/decoding                 |
| <code>cipher.sage</code>   | <code>import crypto.cipher</code>   | XOR cipher, RC4 stream cipher, PKCS#7 padding, CBC/CTR mode helpers |
| <code>rand.sage</code>     | <code>import crypto.rand</code>     | xoshiro256** PRNG, UUID v4, random bytes/strings/hex, shuffle       |
| <code>password.sage</code> | <code>import crypto.password</code> | PBKDF2-HMAC key derivation, password hashing/verification           |

### 9.13 Machine Learning Libraries

SageLang ships with 10 PyTorch-style machine learning modules in `lib/ml/`:

| Module                      | Import                           | Purpose                                                                                             |
|-----------------------------|----------------------------------|-----------------------------------------------------------------------------------------------------|
| <code>tensor.sage</code>    | <code>import ml.tensor</code>    | N-dimensional tensors, element-wise ops, matmul, reductions, activations                            |
| <code>nn.sage</code>        | <code>import ml.nn</code>        | Neural network layers (Linear, ReLU, Sigmoid, Dropout), Sequential model                            |
| <code>optim.sage</code>     | <code>import ml.optim</code>     | SGD (momentum), Adam optimizer, learning rate schedulers                                            |
| <code>loss.sage</code>      | <code>import ml.loss</code>      | MSE, cross-entropy, Huber, L1, hinge, KL divergence                                                 |
| <code>data.sage</code>      | <code>import ml.data</code>      | Dataset/DataLoader, batching, normalization, train/test split                                       |
| <code>debug.sage</code>     | <code>import ml.debug</code>     | Weight stats, histograms, activation analysis, gradient checking, training diagnostics              |
| <code>viz.sage</code>       | <code>import ml.viz</code>       | SVG chart generation (loss curves, weight distributions, attention heatmaps, architecture diagrams) |
| <code>monitor.sage</code>   | <code>import ml.monitor</code>   | Live training monitor, progress bars, memory snapshots, throughput, checkpoints                     |
| <code>gpu_accel.sage</code> | <code>import ml.gpu_accel</code> | GPU offload for tensor ops and training; bridges ML workloads to the Vulkan/OpenGL GPU backend      |
| <code>npu.sage</code>       | <code>import ml.npu</code>       | NPU backend for Qualcomm Hexagon (SNPE), Samsung Exynos (ONE), NNAPI, and ARM NEON SIMD fallback    |

**Native ML Backend (`ml_native`)** `ml_native` is a built-in native C module (not a `lib/ml/` Sage file) that exposes 24 high-performance functions directly from `src/c/ml_backend.c`:

`matmul`, `add`, `scale`, `relu`, `gelu`, `silu`, `sigmoid`, `softmax`, `layer_norm`, `rms_norm`, `cross_entropy`, `adam_update`, `clip_grad`, `benchmark`, `train_step`, `forward_pass`, `load_weights`, `gpu_available`, `set_gpu_threshold`, `cpu_count`, `auto_parallel`, `set_threads`, `set_parallel_threshold`, `get_threads`

#### Backpropagation and training:

- `ml_native.train_step()` — C-level combined forward pass + backward pass + SGD weight update for transformer training. Avoids Python/Sage overhead in the hot loop.
- `ml_native.forward_pass()` — Runs inference using the same computation graph as `train_step`, ensuring numerical consistency between training and evaluation.
- `ml_native.load_weights(path)` — Loads trained weights from a binary file via native C parser; no Sage I/O overhead.

**Standalone C trainer:** `make train-c` builds a standalone C training binary (no Sage runtime required) that uses `ml_backend.c` directly. Auto-detects cuBLAS GPU acceleration and ARM NEON SIMD; also works on mobile via Termux + proot ARM64. Additional targets: `make train-sage` (Sage interpreter), `make chatbot-c` / `make chatbot-llvm` (compile chatbots), `make sl-tq-chat` (SL-TQ-LLM generative chatbot).

## 9.14 CUDA Libraries

SageLang ships with 4 CUDA abstraction modules in `lib/cuda/`:

| Module                   | Import                          | Purpose                                                       |
|--------------------------|---------------------------------|---------------------------------------------------------------|
| <code>device.sage</code> | <code>import cuda.device</code> | GPU device descriptors, compute capability, feature detection |
| <code>memory.sage</code> | <code>import cuda.memory</code> | GPU memory allocation, typed tensors, memory pools            |
| <code>kernel.sage</code> | <code>import cuda.kernel</code> | Kernel definition, launch parameters, occupancy analysis      |
| <code>stream.sage</code> | <code>import cuda.stream</code> | CUDA streams, events, multi-stream execution plans            |

### 9.15 Standard Library (`lib/std/`)

SageLang ships with 23 general-purpose standard library modules in `lib/std/`:

#### Core utilities:

| Module                     | Import                           | Purpose                                                                                                                                                                    |
|----------------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>regex.sage</code>    | <code>import std.regex</code>    | Regular expression engine ( <code>.</code> , <code>*</code> , <code>+</code> , <code>?</code> , <code>[]</code> , <code>\\d</code> , <code>\\w</code> , <code>\\s</code> ) |
| <code>datetime.sage</code> | <code>import std.datetime</code> | Date/time creation, ISO 8601, arithmetic, comparison                                                                                                                       |
| <code>log.sage</code>      | <code>import std.log</code>      | Structured logging (TRACE-FATAL), handlers, child loggers                                                                                                                  |
| <code>argparse.sage</code> | <code>import std.argparse</code> | CLI argument parser (flags, options, positionals)                                                                                                                          |
| <code>compress.sage</code> | <code>import std.compress</code> | RLE, LZ77, delta encoding/decoding                                                                                                                                         |
| <code>process.sage</code>  | <code>import std.process</code>  | Environment, path manipulation, exit codes, timers                                                                                                                         |
| <code>unicode.sage</code>  | <code>import std.unicode</code>  | UTF-8, character classification, case conversion, trim                                                                                                                     |
| <code>fmt.sage</code>      | <code>import std.fmt</code>      | Number/string formatting, templates, table output                                                                                                                          |
| <code>testing.sage</code>  | <code>import std.testing</code>  | Test runner, assertions, benchmarks                                                                                                                                        |
| <code>db.sage</code>       | <code>import std.db</code>       | In-memory DB (CRUD, joins, aggregation, pagination)                                                                                                                        |
| <code>signal.sage</code>   | <code>import std.signal</code>   | Event bus (pub/sub), atexit handlers                                                                                                                                       |

#### Type system extensions:

| Module                  | Import                        | Purpose                                                   |
|-------------------------|-------------------------------|-----------------------------------------------------------|
| <code>enum.sage</code>  | <code>import std.enum</code>  | Enums, tagged unions, Result (Ok/Err), Option (Some/None) |
| <code>trait.sage</code> | <code>import std.trait</code> | Interface/trait system, behavioral contracts              |

#### Concurrency primitives:

| Module                       | Import                             | Purpose                                            |
|------------------------------|------------------------------------|----------------------------------------------------|
| <code>channel.sage</code>    | <code>import std.channel</code>    | Go-style channels, send/recv/select, fan-in/out    |
| <code>threadpool.sage</code> | <code>import std.threadpool</code> | Work queue, parallel map, futures/promises         |
| <code>atomic.sage</code>     | <code>import std.atomic</code>     | Atomic integers, CAS, spin locks, counters         |
| <code>rwlock.sage</code>     | <code>import std.rwlock</code>     | Read-write locks, scoped lock helpers              |
| <code>condvar.sage</code>    | <code>import std.condvar</code>    | Condition variables, barriers, latches, semaphores |

#### Developer tooling:

| Module                     | Import                           | Purpose                                               |
|----------------------------|----------------------------------|-------------------------------------------------------|
| <code>debug.sage</code>    | <code>import std.debug</code>    | Value inspection, trace logging, watch expressions    |
| <code>profiler.sage</code> | <code>import std.profiler</code> | Hierarchical timing, hotspots, benchmark runner       |
| <code>docgen.sage</code>   | <code>import std.docgen</code>   | Doc extraction from comments, markdown generation     |
| <code>build.sage</code>    | <code>import std.build</code>    | Project config, dependencies, semver, build targets   |
| <code>interop.sage</code>  | <code>import std.interop</code>  | FFI helpers, C types, pack/unpack, platform detection |

## 9.16 LLM / Neural Network Libraries

SageLang ships with 19 LLM/neural network modules in `lib/llm/` for building and training language models:

**Backpropagation note:** For performance-critical training loops, use `ml_native.train_step()` (C-level forward+backward+SGD), `ml_native.forward_pass()` (inference, same computation graph), and `ml_native.load_weights(path)` (native weight loading) instead of the pure-Sage `llm.train` module. See Section 9.13 Native ML Backend.

| Module                                                       | Import                                                                   | Purpose                                                                                           |
|--------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <code>config.sage</code>                                     | <code>import llm.config</code>                                           | Model configs (tiny to Llama-13B), param counting, memory estimation                              |
| <code>tokenizer.sage</code>                                  | <code>import llm.tokenizer</code>                                        | Character, word, and BPE tokenizers                                                               |
| <code>embedding.sage</code>                                  | <code>import llm.embedding</code>                                        | Token embeddings, sinusoidal/learned/RoPE positional encodings                                    |
| <code>attention.sage</code><br><code>transformer.sage</code> | <code>import llm.attention</code><br><code>import llm.transformer</code> | Multi-head self-attention, KV cache<br>Transformer blocks, LayerNorm/RMSNorm, FFN, model assembly |
| <code>generate.sage</code>                                   | <code>import llm.generate</code>                                         | Text generation (greedy, top-k, top-p, beam search)                                               |
| <code>train.sage</code>                                      | <code>import llm.train</code>                                            | Training loops, LR schedules, cross-entropy, perplexity                                           |

| Module                         | Import                               | Purpose                                                                                                                                                                                                                     |
|--------------------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>agent.sage</code>        | <code>import llm.agent</code>        | Agentic framework (tools, CoT, memory, planning, multi-agent)                                                                                                                                                               |
| <code>prompt.sage</code>       | <code>import llm.prompt</code>       | Chat formatting (ChatML/Llama/Alpaca), templates, few-shot                                                                                                                                                                  |
| <code>lora.sage</code>         | <code>import llm.lora</code>         | LoRA fine-tuning adapters, merge-back                                                                                                                                                                                       |
| <code>quantize.sage</code>     | <code>import llm.quantize</code>     | Int8/int4 quantization, error analysis                                                                                                                                                                                      |
| <code>engram.sage</code>       | <code>import llm.engram</code>       | Persistent neural memory (working/episodic/semantic/procedural)                                                                                                                                                             |
| <code>rag.sage</code>          | <code>import llm.rag</code>          | Document chunking, keyword retrieval, context assembly, summarization                                                                                                                                                       |
| <code>dpo.sage</code>          | <code>import llm.dpo</code>          | Direct Preference Optimization, ORPO, preference pairs, reward models                                                                                                                                                       |
| <code>gguf.sage</code>         | <code>import llm.gguf</code>         | GGUF v3 export for Ollama/llama.cpp, Modelfile gen, quantization                                                                                                                                                            |
| <code>gguf_import.sage</code>  | <code>import llm.gguf_import</code>  | Import GGUF models from Ollama into Sage; converts weights to native tensor format                                                                                                                                          |
| <code>turboquant.sage</code>   | <code>import llm.turboquant</code>   | TurboQuant near-optimal vector quantization (ICLR 2026): two-stage PolarQuant (random rotation + MSE-optimal scalar quantization) + QJL (1-bit residual correction); KV cache compression at 3-bit with 6x memory reduction |
| <code>autoresearch.sage</code> | <code>import llm.autoresearch</code> | Karpathy-style autonomous research agent; ratchet loop (propose → train → evaluate → accept/reject); built-in scale/choice/perturb strategies; research journal; multi-agent session merging                                |
| <code>evolve.sage</code>       | <code>import llm.evolve</code>       | Self-evolving neural architecture; progressive growth from seed (64d/1L/98K params) to ancient (512d/8L/67M params); auto-plateau detection; weight padding for width growth; identity-init for depth growth                |

### 9.17 Agent Framework (lib/agent/)

| Module                  | Import                          | Purpose                                                      |
|-------------------------|---------------------------------|--------------------------------------------------------------|
| <code>core.sage</code>  | <code>import agent.core</code>  | ReAct agent loop, tool dispatch, scratchpad, prompt building |
| <code>tools.sage</code> | <code>import agent.tools</code> | Pre-built tools (file I/O, code analysis, search, system)    |

| Module                            | Import                                    | Purpose                                                        |
|-----------------------------------|-------------------------------------------|----------------------------------------------------------------|
| <code>planner.sage</code>         | <code>import agent.planner</code>         | Task decomposition with dependency DAG, auto-execution         |
| <code>router.sage</code>          | <code>import agent.router</code>          | Multi-agent orchestrator, capability routing, pipelines        |
| <code>supervisor.sage</code>      | <code>import agent.supervisor</code>      | Supervisor-Worker control plane, workflow engine, retries      |
| <code>critic.sage</code>          | <code>import agent.critic</code>          | Verification loops, rule/LLM validators, composite checks      |
| <code>schema.sage</code>          | <code>import agent.schema</code>          | Typed tool interfaces, parameter validation, bounded execution |
| <code>trace.sage</code>           | <code>import agent.trace</code>           | SFT trace recording, training data generation                  |
| <code>grammar.sage</code>         | <code>import agent.grammar</code>         | Grammar-constrained decoding, tool call/JSON/Sage validation   |
| <code>sandbox.sage</code>         | <code>import agent.sandbox</code>         | Program-aided reasoning, code execution, math eval             |
| <code>tot.sage</code>             | <code>import agent.tot</code>             | Tree of Thoughts, BFS/best-first search, state rollbacks       |
| <code>semantic_router.sage</code> | <code>import agent.semantic_router</code> | Fast command dispatch, keyword routing, LLM bypass             |

### 9.18 Chatbot Framework (`lib/chat/`)

| Module                    | Import                           | Purpose                                                     |
|---------------------------|----------------------------------|-------------------------------------------------------------|
| <code>bot.sage</code>     | <code>import chat.bot</code>     | Conversation management, intents, middleware, LLM responses |
| <code>session.sage</code> | <code>import chat.session</code> | Multi-session store, history, export (text/JSON)            |
| <code>persona.sage</code> | <code>import chat.persona</code> | Pre-built personas (SageDev, Teacher, Debugger, etc.)       |

### 9.19 UI Widget Library (`lib/graphics/ui.sage`)

Immediate-mode GPU UI system for building application interfaces:

```
import gpu
import graphics.ui

gpu.init_windowed("App", 800, 600, "My App", false)
let ctx = ui.ui_create()

while not gpu.window_should_close():
    gpu.poll_events()
    ui.ui_begin_frame(ctx)

    ui.ui_panel(ctx, 10, 10, 300, 400, "Settings")
    if ui.ui_button(ctx, 20, 50, 120, 30, "Apply"):
        print "Applied!"

    let volume = ui.ui_slider(ctx, 20, 100, 260, "Volume", 0.75)
    let dark = ui.ui_checkbox(ctx, 20, 140, "Dark Mode", true)
```

```
ui.ui_end_frame(ctx)
```

Available widgets:

| Widget     | Function                                                     | Returns                |
|------------|--------------------------------------------------------------|------------------------|
| Label      | <code>ui_label(ctx, x, y, text)</code>                       | nil                    |
| Button     | <code>ui_button(ctx, x, y, w, h, label)</code>               | bool (clicked)         |
| Panel      | <code>ui_panel(ctx, x, y, w, h, title)</code>                | nil                    |
| Window     | <code>ui_window(ctx, x, y, w, h, title)</code>               | dict (content area)    |
| Checkbox   | <code>ui_checkbox(ctx, x, y, label, checked)</code>          | bool (new state)       |
| Slider     | <code>ui_slider(ctx, x, y, w, label, value)</code>           | number (0.0-1.0)       |
| Scrollbar  | <code>ui_scrollbar_v(ctx, x, y, h, content_h, scroll)</code> | number (0.0-1.0)       |
| Menu       | <code>ui_menu_button(ctx, x, y, w, h, label, items)</code>   | int (item index or -1) |
| Text Input | <code>ui_text_input(ctx, x, y, w, label, text)</code>        | string                 |
| Progress   | <code>ui_progress(ctx, x, y, w, h, value, label)</code>      | nil                    |
| Separator  | <code>ui_separator(ctx, x, y, w)</code>                      | nil                    |
| Tooltip    | <code>ui_tooltip(ctx, text)</code>                           | nil                    |

Theming: `ctx["theme"]` is a dict with 20+ configurable colors (`bg`, `accent`, `text`, `border`, etc.) and sizes (`padding`, `font_size`, `title_height`, etc.). Call `ui.ui_default_theme()` for defaults.

## Part 13: Self-Hosting (Phase 13)

SageLang is **self-hosted**: the lexer, parser, and interpreter have been ported from C to Sage itself. The self-hosted pipeline can execute `.sage` programs using only the C interpreter as a bootstrap host.

### 13.1 Architecture

The self-hosted interpreter lives in `src/sage/` and consists of:

| File                          | Description                                | Size        |
|-------------------------------|--------------------------------------------|-------------|
| <code>token.sage</code>       | Token type constants                       | ~50 lines   |
| <code>lexer.sage</code>       | Tokenizer with indentation tracking        | ~300 lines  |
| <code>ast.sage</code>         | AST node constructors (dict-based)         | ~100 lines  |
| <code>parser.sage</code>      | Recursive descent parser                   | ~700 lines  |
| <code>interpreter.sage</code> | Tree-walking evaluator with module imports | ~1050 lines |
| <code>errors.sage</code>      | Rich error reporting (Rust/Elm-style)      | ~200 lines  |
| <code>environment.sage</code> | Dict-based scope/environment               | ~35 lines   |
| <code>sage.sage</code>        | Full CLI entry point                       | ~200 lines  |

## 13.2 Running Self-Hosted Code

```
cd src/sage
../../sage sage.sage program.sage
```

The bootstrap reads a `.sage` file, tokenizes it, parses it to an AST, and evaluates it — all in Sage running on the C interpreter.

## 13.3 Key Design Decisions

**Dict-based value representation:** While SageLang recently added native C-like enums (Phase 1.7), the self-hosted interpreter relies on dictionary-based representations for its AST nodes because it was designed before these features existed and for simplicity. All AST nodes, functions, classes, and instances are represented as dicts with an `__interp_type` field:

```
# A function value
let fn = {}
fn["__interp_type"] = "function"
fn["name"] = "add"
fn["params"] = ["a", "b"]
fn["body"] = body_ast
```

**Control flow signals:** Return, break, and continue are implemented as dict values:

```
# return 42 → {"kind": "return", "value": 42}
# break    → {"kind": "break"}
# continue → {"kind": "continue"}
# normal   → {"kind": "normal"}
```

**GC must be disabled:** The self-hosted interpreter creates many dict allocations, which can trigger GC segfaults. Always start with `gc_disable()`.

## 13.4 Feature Coverage

The self-hosted interpreter supports (~70% feature parity with C):

**Fully implemented:** arithmetic, variables, control flow (if/elif/else, while, for), functions with closures, recursion, classes with inheritance, arrays, dicts, tuples, strings, slicing, indexing, try/catch/finally, break/continue, raise, module imports (`import X`, `import X as Y`, `from X import a, b`), property access/set, all bitwise operators (`&` | `^` `~` `<<` `>>`), `match/case/default` pattern matching, `defer` statements (LIFO cleanup on scope exit), generators/yield (eager collection via `next()`), 35+ builtins.

For self-hosted LLVM codegen specifically (`src/sage/llvm_backend.sage`), `from X import Y` constant imports are resolved during code generation for foldable top-level `let` values (including `as` aliases), matching the C LLVM backend behavior for cross-module constants.

**Delegated to host runtime:** GC control (`gc_collect`, `gc_enable`, `gc_disable`, `gc_stats`), FFI (`ffi_open`, `ffi_close`, `ffi_call`, `ffi_sym`), memory access (`mem_alloc`, `mem_free`, `mem_read`, `mem_write`, `mem_size`, `addressof`), networking (via host `import` of native modules).

**Stub/partial:** `async proc` (registered with `is_async` flag, executes synchronously), `await` (evaluates expression directly).

**Not implemented:** true coroutine-based generators (uses eager collection instead), actual async threading in self-hosted path, GPU module (uses host `import gpu` directly).

**Safety:** while loop iteration limit (1M), recursion depth limit (500), rich error messages with source context via `errors.sage`.

## 13.5 Test Suite

The self-hosted tree includes core parser/interpreter suites plus additional tooling, optimizer, backend, compiler, error-reporting, LSP, and CLI tests. The foundational bootstrap suites are:

| Category                           | Tests | Coverage                                                                       |
|------------------------------------|-------|--------------------------------------------------------------------------------|
| <code>test_lexer.sage</code>       | 13    | Token types, indentation, keywords                                             |
| <code>test_parser.sage</code>      | 130   | All AST node types, operator precedence                                        |
| <code>test_interpreter.sage</code> | 18    | Evaluation, scoping, closures, classes                                         |
| <code>test_bootstrap.sage</code>   | 18    | End-to-end: source $\rightarrow$ tokens $\rightarrow$ AST $\rightarrow$ result |

The full test suite (interpreter + compiler + self-hosted tooling) totals **2060+ tests**.

---

## Part 10: Native Standard Library Modules (Phase 11)

SageLang provides built-in native modules that are pre-loaded into the module cache and available via `import` without any file on disk.

### 10.1 Math Module

```
import math
```

```
print math.sqrt(16)      # 4
print math.sin(math.pi) # ~0
print math.floor(3.7)    # 3
print math.ceil(3.2)     # 4
print math.abs(-5)       # 5
print math.pow(2, 10)    # 1024
print math.log(math.e)   # 1
math.printm("10 + 20") # Show work for 10 + 20
```

Available functions: `sqrt`, `sin`, `cos`, `tan`, `floor`, `ceil`, `abs`, `pow`, `log`, `printm` Constants: `pi` (3.14159...), `e` (2.71828...)

### 10.2 IO Module

```
import io
```

```
io.writefile("test.txt", "Hello, World!")
let content = io.readfile("test.txt")
print content # Hello, World!
```

```
io.appendfile("test.txt", "\nLine 2")
print io.exists("test.txt") # true
```

```
io.rename("test.txt", "renamed.txt")
io.remove("renamed.txt")
```

```
# Binary I/O
let buf = bytes("Hello\n")
```

```
io.writebytes("data.bin", buf)      # Write Bytes value (also accepts Array)
let r = io.readbytes("data.bin")    # Read as Bytes value
io.appendbytes("log.bin", buf)      # Append Bytes value
```

Available functions: `readfile`, `writefile`, `appendfile`, `exists`, `remove`, `rename`, `readbytes`, `writebytes`, `appendbytes`

*Note: `io.readbytes` returns a *Bytes* value (byte buffer), and `io.writebytes`/`io.appendbytes` accept either *Bytes* or *Array* values.*

### 10.3 String Module

```
import string
```

```
print string.char(65)           # A
print string.ord("A")           # 65
print string.startswith("hello", "he") # true
print string.endswith("hello", "lo")   # true
print string.contains("hello", "ell")   # true
print string.repeat("ab", 3)            # ababab
print string.reverse("hello")           # olleh
```

Available functions: `char`, `ord`, `startswith`, `endswith`, `contains`, `repeat`, `reverse`

### 10.4 Sys Module

```
import sys
```

```
print sys.platform() # linux, darwin, or windows
print sys.version()  # Sage version string
let path = sys.env("HOME")
print path
```

Available functions: `args`, `exit`, `platform`, `version`, `env`, `setenv`

### 10.5 FAT Module

The native `fat` module provides early FAT filesystem parsing helpers for image inspection and kernel/boot tooling.

```
import os.fat
import io
```

```
let boot = io.readbytes("disk.img")
let info = fat.parse_boot_sector(boot)
```

```
print info["fat_type"]           # FAT8 / FAT12 / FAT16 / FAT32
print info["cluster_count"]
print fat.cluster_to_lba(info, 2) # First data cluster -> LBA/sector index
```

Available functions: - `parse_boot_sector(bytes_array)` - `probe(path)` - `cluster_to_lba(info, cluster)` - `fat_entry_offset(info, cluster)`

Constants: - `FAT8`, `FAT12`, `FAT16`, `FAT32`

Current scope: - Boot sector parsing, FAT type detection, and layout math for FAT8/12/16/32. - Intended as a foundation for follow-up directory walking, FAT chain traversal, and read/write support.

## Part 11: Concurrency and Async/Await (Phase 11)

### 11.1 Thread Module

The `thread` module provides low-level threading primitives backed by `pthread`s.

```
import thread

proc worker(name):
  print name
  return 42

# Spawn a thread
let t = thread.spawn(worker, "hello")

# Wait for result
let result = thread.join(t)
print result # 42
```

#### Thread Functions

| Function                                 | Description                                                       |
|------------------------------------------|-------------------------------------------------------------------|
| <code>thread.spawn(proc, args...)</code> | Spawn a new thread running <code>proc</code> with given arguments |
| <code>thread.join(t)</code>              | Block until thread <code>t</code> completes, return its result    |
| <code>thread.mutex()</code>              | Create a new mutex                                                |
| <code>thread.lock(m)</code>              | Acquire mutex <code>m</code> (blocks if held)                     |
| <code>thread.unlock(m)</code>            | Release mutex <code>m</code>                                      |
| <code>thread.sleep(ms)</code>            | Sleep for <code>ms</code> milliseconds                            |
| <code>thread.id()</code>                 | Return current thread identifier                                  |

#### Mutex Example

```
import thread

let m = thread.mutex()

proc critical_section():
  thread.lock(m)
  # ... protected code ...
  thread.unlock(m)
```

### 11.2 Async/Await

The `async proc` keyword declares a procedure that runs asynchronously. Calling it spawns a background thread and returns a thread handle. Use `await` to retrieve the result.

```
async proc compute(x):
  return x * x

# Calling an async proc spawns a thread
let future = compute(5)

# await blocks until the thread completes
let result = await future
print result # 25
```

## Parallel Execution

```
async proc slow_add(a, b):  
    return a + b  
  
# Launch two computations in parallel  
let f1 = slow_add(1, 2)  
let f2 = slow_add(3, 4)  
  
# Await both results  
let r1 = await f1  
let r2 = await f2  
print r1 + r2 # 10
```

## How It Works

1. `async proc` is parsed as `STMT_ASYNC_PROC` and sets `is_async = 1` on the `FunctionValue`
2. When called, the interpreter pre-evaluates arguments and spawns a thread via `thread_spawn_native`
3. The call returns a `VAL_THREAD` value (a thread handle)
4. `await` on a `VAL_THREAD` calls `pthread_join` and returns the thread's result value

## 11.3 GC Thread Safety

The garbage collector is protected by a pthread mutex. All GC operations (allocation, collection, marking, sweeping) acquire the lock, ensuring safe concurrent allocation from multiple threads.

---

---

## Part 12: Developer Tooling (Phase 12)

SageLang includes a complete set of developer tools for interactive development, code quality, and editor integration.

### 12.1 REPL (Read-Eval-Print Loop)

Launch the interactive REPL by running `sage` with no arguments or with the `--repl` flag:

```
sage  
sage --repl
```

The REPL supports multi-line blocks (indented code is automatically continued), error recovery (errors are displayed without exiting), and built-in commands.

#### Built-in Commands:

| Command                         | Description                                                 |
|---------------------------------|-------------------------------------------------------------|
| <code>:help</code>              | Show REPL usage information                                 |
| <code>:quit / :exit</code>      | Exit the REPL                                               |
| <code>:vars [prefix]</code>     | List current REPL bindings, optionally filtered by prefix   |
| <code>:type &lt;expr&gt;</code> | Evaluate an expression and print its runtime type and value |
| <code>:load &lt;file&gt;</code> | Execute a Sage file inside the current REPL session         |
| <code>:reset</code>             | Reset the REPL environment and module cache                 |
| <code>:pwd</code>               | Print the current working directory                         |
| <code>:cd &lt;dir&gt;</code>    | Change the current working directory                        |
| <code>:gc</code>                | Run garbage collection and print GC statistics              |

| Command             | Description                              |
|---------------------|------------------------------------------|
| <code>:stats</code> | Show GC stats, stack depth, and CPU time |

#### Example Session:

```
sage> let x = 10
sage> proc double(n):
....>     return n * 2
sage> print double(x)
20
sage> :quit
```

## 12.2 Code Formatter

The formatter normalizes indentation, spacing, and blank lines for consistent code style.

```
sage fmt program.sage           # Format file in place
sage fmt --check program.sage   # Check formatting (exit code 1 if changes needed)
```

#### Example:

```
$ sage fmt --check messy.sage
messy.sage: needs formatting
$ sage fmt messy.sage
Formatted: messy.sage
$ sage fmt --check messy.sage
messy.sage: already formatted
```

## 12.3 Linter

The linter performs static analysis with 13 rules across three categories:

```
sage lint program.sage
```

#### Rule Categories:

| Category | Rules     | Description                       |
|----------|-----------|-----------------------------------|
| Errors   | E001-E003 | Syntax and structural errors      |
| Warnings | W001-W005 | Potential bugs and bad practices  |
| Style    | S001-S005 | Code style and naming conventions |

**Notable Rules:** - [W001], [W002]: Issued for unused variables and variables that shadow existing declarations in scope. - [W004]: Warns about empty blocks (e.g., when a line ending in a colon `:` is followed by an empty or improperly indented section). - [S003]: Enforces documentation conventions; top-level `proc` declarations must be immediately preceded by a `##` style docstring (comment).

#### Example Output:

```
program.sage:5: W003 unused variable 'temp'
program.sage:12: S002 line too long (exceeds 100 characters)
program.sage:20: E001 undefined variable 'foo'
```

## 12.4 Syntax Highlighting

SageLang provides editor support via TextMate grammars:

- **TextMate grammar:** `editors/sage.tmLanguage.json` works with any TextMate-compatible editor (VSCode, Sublime Text, etc.)
- **VSCode extension:** `editors/vscode/` provides full language support including syntax highlighting and language configuration

### Installing the VSCode Extension:

1. Copy or symlink `editors/vscode/` into `~/.vscode/extensions/sage-lang/`
2. Restart VSCode
3. Open any `.sage` file to see syntax highlighting

## 12.5 Language Server Protocol (LSP)

The LSP server provides IDE-like features for any editor that supports the Language Server Protocol.

```
sage --lsp           # Start LSP server via main binary
sage-lsp             # Standalone LSP server binary
```

### Capabilities:

| Feature     | Description                                             |
|-------------|---------------------------------------------------------|
| Diagnostics | Real-time error and warning reporting on file save      |
| Completion  | Keyword and symbol completions as you type              |
| Hover       | Type information and documentation on hover             |
| Formatting  | Format-on-save via <code>textDocument/formatting</code> |

### Editor Configuration (VSCode `settings.json`):

```
{
  "sage.lsp.path": "/path/to/sage-lsp"
}
```

## Example Programs

### Hello World:

```
print "Hello, World"
```

### Factorial:

```
proc factorial(n):
  if n <= 1:
    return 1
  return n * factorial(n - 1)
```

```
print factorial(5) # 120
```

### Fibonacci:

```
proc fib(n):
  if n <= 1:
    return n
  return fib(n - 1) + fib(n - 2)
```

```
for i in range(0, 10):
    print fib(i)
```

### Object-Oriented:

```
class Rectangle:
    proc init(width, height):
        self.width = width
        self.height = height

    proc area():
        return self.width * self.height

let rect = Rectangle(5, 3)
print rect.area() # 15
```

## Part 14: Networking (Phase 14)

SageLang provides four native networking modules backed by libcurl and OpenSSL. These are implemented in C (`src/net.c`) and registered as native modules.

### 14.1 Socket Module

Low-level POSIX socket operations with constants for address families and socket types.

```
import socket

# TCP client
let sock = socket.create(socket.AF_INET, socket.SOCK_STREAM, 0)
socket.connect(sock, "example.com", 80)
socket.send(sock, "GET / HTTP/1.0" + chr(13) + chr(10) + chr(13) + chr(10))
let response = socket.recv(sock, 4096)
print response
socket.close(sock)

# DNS resolution
let ip = socket.resolve("example.com")
print ip # "3.8.0.34"

# UDP
let udp = socket.create(socket.AF_INET, socket.SOCK_DGRAM, 0)
socket.sendto(udp, "hello", "3.8.0.1", 9999)
socket.close(udp)
```

**Constants:** AF\_INET, AF\_INET6, SOCK\_STREAM, SOCK\_DGRAM, SOCK\_RAW, IPPROTO\_TCP, IPPROTO\_UDP

**Functions** (15): create, bind, listen, accept, connect, send, recv, sendto, recvfrom, close, setopt, poll, resolve, getpeername, nonblock

### 14.2 TCP Module

High-level TCP with automatic buffering and convenience functions.

```
import tcp

# Client
let conn = tcp.connect("example.com", 80)
tcp.sendall(conn, "GET / HTTP/1.0" + chr(13) + chr(10) + chr(13) + chr(10))
```

```
let line = tcp.recvline(conn)
print line
tcp.close(conn)
```

```
# Server
let server = tcp.listen("3.8.0.0", 8080, 5)
let client = tcp.accept(server)
tcp.send(client, "Hello from Sage!")
tcp.close(client)
tcp.close(server)
```

**Functions** (9): connect, listen, accept, send, recv, sendall, recvall, recvline, close

### 14.3 HTTP Module

HTTP/HTTPS client via libcurl. All request functions return a dict with **status**, **body**, and **headers** keys.

```
import http
```

```
# Simple GET
let resp = http.get("https://httpbin.org/get")
print resp["status"]    # 200
print resp["body"]

# POST with options
let opts = {"timeout": 30, "headers": {"Content-Type": "application/json"}}
let resp2 = http.post("https://httpbin.org/post", "{}", opts)

# Download file
http.download("https://example.com/file.txt", "/tmp/file.txt")

# URL encoding
print http.escape("hello world")    # "hello%20world"
print http.unescape("hello%20world") # "hello world"
```

**Functions** (9): get, post, put, delete, patch, head, download, escape, unescape

**Options dict keys:** timeout (seconds), follow (bool, follow redirects), verify (bool, SSL verification), user\_agent (string), headers (dict of header name→value pairs), cainfo (CA certificate path)

### 14.4 SSL Module

OpenSSL TLS/SSL bindings for encrypted socket communication.

```
import socket
import ssl
```

```
let sock = socket.create(socket.AF_INET, socket.SOCK_STREAM, 0)
socket.connect(sock, "example.com", 443)
```

```
let ctx = ssl.context()
let s = ssl.wrap(ctx, sock)
ssl.connect(s)
ssl.send(s, "GET / HTTP/1.1" + chr(13) + chr(10) + "Host: example.com" + chr(13) + chr(10) + chr(13) + chr(10))
let data = ssl.recv(s, 4096)
print data
```

```
ssl.shutdown(s)
ssl.free(s)
ssl.free_context(ctx)
socket.close(sock)
```

**Functions (13):** context, load\_cert, wrap, connect, accept, send, recv, shutdown, free, free\_context, error, peer\_cert, set\_verify

## 14.5 Important Notes

- **No escape sequences:** Sage strings are raw. Use `chr(13) + chr(10)` for CRLF, `chr(34)` for double-quote, `chr(92)` for backslash.
- **HTTP module** handles HTTPS automatically via libcurl — no need for manual SSL setup for HTTP requests.
- **Build requirement:** libcurl and openssl development libraries must be installed.

## Part 15: JSON Library (cJSON Port)

SageLang includes a complete 1:1 port of Dave Gamble's cJSON library in `lib/json.sage` (~1,050 lines). It uses a linked-list tree structure mirroring the C original.

### 15.1 Basic Usage

```
gc_disable() # Required for class-heavy code
```

```
from json import cJSON_Parse, cJSON_Print, cJSON_PrintUnformatted
from json import cJSON_GetObjectItem, cJSON_GetStringValue, cJSON_GetNumberValue
from json import cJSON_ToSage, cJSON_Delete
```

```
# Parse JSON string (use chr(34) for double-quotes in Sage)
```

```
let json_str = "{" + chr(34) + "name" + chr(34) + ":" + chr(34) + "Alice" + chr(34) + "," + chr(34) + "age" + chr(34) + ":30" + chr(34) + "}"
let root = cJSON_Parse(json_str)
```

```
# Query values
```

```
let name = cJSON_GetStringValue(cJSON_GetObjectItem(root, "name"))
let age = cJSON_GetNumberValue(cJSON_GetObjectItem(root, "age"))
print name # Alice
print age  # 30
```

```
# Print JSON
```

```
print cJSON_Print(root) # Formatted (pretty-printed)
print cJSON_PrintUnformatted(root) # Compact
```

```
# Convert to native Sage dict
```

```
let native = cJSON_ToSage(root)
print native["name"] # Alice
```

```
cJSON_Delete(root) # No-op in GC language, included for API compatibility
```

### 15.2 Creating JSON

```
gc_disable()
from json import cJSON_CreateObject, cJSON_CreateArray
from json import cJSON_AddStringToObject, cJSON_AddNumberToObject
```

```

from json import cJSON_AddBoolToObject, cJSON_AddNullToObject
from json import cJSON_AddArrayToObject, cJSON_AddItemToArray
from json import cJSON_CreateNumber, cJSON_PrintUnformatted

let obj = cJSON_CreateObject()
cJSON_AddStringToObject(obj, "name", "Bob")
cJSON_AddNumberToObject(obj, "age", 25)
cJSON_AddBoolToObject(obj, "active", true)
cJSON_AddNullToObject(obj, "deleted_at")

let tags = cJSON_AddArrayToObject(obj, "tags")
cJSON_AddItemToArray(tags, cJSON_CreateNumber(1))
cJSON_AddItemToArray(tags, cJSON_CreateNumber(2))

print cJSON_PrintUnformatted(obj)
# {"name":"Bob","age":25,"active":true,"deleted_at":null,"tags":[1,2]}

```

### 15.3 Sage-Native Conversion

```

gc_disable()
from json import cJSON_FromSage, cJSON_ToSage, cJSON_Print

# Native Sage value → cJSON tree
let data = {"users": [{"name": "Alice"}, {"name": "Bob"}], "count": 2}
let tree = cJSON_FromSage(data)
print cJSON_Print(tree)

# cJSON tree → native Sage value
let back = cJSON_ToSage(tree)
print back["users"][0]["name"] # Alice

```

### 15.4 API Reference

| Category           | Functions                                                                                                                              |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Parsing</b>     | cJSON_Parse, cJSON_ParseWithLength,<br>cJSON_GetErrorPtr                                                                               |
| <b>Printing</b>    | cJSON_Print, cJSON_PrintUnformatted,<br>cJSON_PrintBuffered                                                                            |
| <b>Creation</b>    | cJSON_CreateNull/True/False/Bool/Number/String/Raw/Array/Ob<br>CreateIntArray/DoubleArray/FloatArray/StringArray                       |
| <b>Query</b>       | cJSON_GetArraySize, GetArrayItem, GetObjectItem,<br>GetObjectItemCaseSensitive, HasObjectItem,<br>GetStringValue, GetNumberValue       |
| <b>Type Checks</b> | cJSON_IsInvalid/False/True/Bool/Null/Number/String/Array/Ob                                                                            |
| <b>Array Ops</b>   | AddItemToArray, InsertItemInArray,<br>DetachItemFromArray, DeleteItemFromArray,<br>ReplaceItemInArray                                  |
| <b>Object Ops</b>  | AddItemToObject/CS,<br>DetachItemFromObject/CaseSensitive,<br>DeleteItemFromObject/CaseSensitive,<br>ReplaceItemInObject/CaseSensitive |
| <b>Helpers</b>     | cJSON_AddNullToObject,<br>AddTrue/False/Bool/Number/String/Raw/Array/ObjectToObject                                                    |

| Category    | Functions                                                                          |
|-------------|------------------------------------------------------------------------------------|
| Utility     | cJSON_Duplicate, Compare, Minify, Delete, SetValuestring, SetNumberHelper, Version |
| Sage Extras | cJSON_ToSage (tree→native), cJSON_FromSage (native→tree)                           |

### 15.5 Important Notes

- **GC must be disabled** (`gc_disable()`) when using `json.sage` — class-heavy code triggers GC seg-faults.
- `cJSON_GetObjectItem` does **case-insensitive** matching (uses `lower()`). Use `cJSON_GetObjectItemCaseSensitive` for exact matching.
- `cJSON_Delete` is a no-op (Sage has GC), included for API compatibility with C cJSON.
- **Type constants:** `cJSON_Invalid=0`, `cJSON_False=1`, `cJSON_True=2`, `cJSON_NULL=4`, `cJSON_Number=8`, `cJSON_String=16`, `cJSON_Array=32`, `cJSON_Object=64`, `cJSON_Raw=128`

## Conclusion

**SageLang** is a comprehensive, well-structured scripting language for systems and embedded programming. Its design combines the approachability of Python with low-level control through direct memory access, FFI, and inline assembly. The phased development approach (Phases 1–18) progresses from core features through advanced topics including OOP, generators, compilation backends (C, LLVM IR, native assembly), concurrency, networking, JSON processing, and a self-hosted interpreter — all fully implemented and integrated.

**Key Takeaways:** - **Lexer + Parser + Interpreter** pipeline is modular and extensible. - **Value system** supports dynamic typing with heap-allocated complex objects and GC. - **Scoped environments** enable closures and lexical scoping. - **Exception handling, generators, and `async/await`** provide modern control flow. - **Module system** enables code reuse with native (C) and Sage library modules. - **Mark-and-sweep GC** manages memory automatically with thread safety. - **Multiple backends:** tree-walking interpreter, C codegen, LLVM IR, native assembly (x86-64, aarch64, rv64, mips). - **Networking:** POSIX sockets, TCP, HTTP/HTTPS (libcurl), SSL/TLS (OpenSSL). - **Self-hosted:** Lexer, parser, and interpreter ported to Sage with full bootstrap. - **Test suite:** interpreter/compiler coverage, JSON coverage, and broad self-hosted suites spanning parsing, tooling, optimization, codegen, compiler, LSP, and CLI behavior.

SageLang offers a practical balance between ease of use and systems-level control, making it ideal for prototyping, education, embedded scripting, and learning language implementation.

## GPU Graphics (Vulkan + OpenGL)

SageLang includes a professional-grade graphics engine supporting both Vulkan and OpenGL 4.5 backends. The library is built in four layers:

1. **Pure C GPU API** (`gpu_api.h/gpu_api.c`) – Backend-agnostic ~100 functions, no interpreter dependency
2. **C native module** (`import gpu`) – Interpreter wrappers for Value-based argument handling
3. **Sage builder libraries** (`lib/vulkan.sage`, `lib/opengl.sage`, `lib/gpu.sage`) – Ergonomic helpers
4. **Engine + UI libraries** (`lib/renderer.sage`, `lib/pbr.sage`, `lib/ui.sage`, etc.) – Application-level systems

Three execution paths share the same GPU API:

- **Interpreter:** `graphics.c` wraps `sgpu_*` with Value extraction
- **LLVM compiled:** `llvm_runtime.c` provides 103 `sage_rt_gpu_*` bridge functions

- **Bytecode VM:** 30 dedicated BC\_OP\_GPU\_\* opcodes for frame-loop hot paths

## Build Requirements

Vulkan and OpenGL support are auto-detected at build time via `pkg-config`. GLFW is required for windowed rendering:

```
# Auto-detect Vulkan + OpenGL (default)
make
```

```
# Force enable/disable
make VULKAN=1      # Force Vulkan
make VULKAN=0      # Disable Vulkan
make OPENGGL=1     # Force OpenGL
make OPENGGL=0     # Disable OpenGL
```

Without the Vulkan SDK, the `gpu` module loads in stub mode – all constants are available, functions return errors gracefully. Use `import graphics.opengl` instead of `import gpu` for the OpenGL 4.5 backend (same API, different initialization).

## Quick Start: Empty Window

```
import gpu

gpu.init_windowed("My App", 800, 600, "Window Title", false)
print gpu.device_name()

let attach = {}
attach["format"] = gpu.FORMAT_SWAPCHAIN
attach["load_op"] = gpu.LOAD_CLEAR
attach["store_op"] = gpu.STORE_STORE
attach["initial_layout"] = gpu.LAYOUT_UNDEFINED
attach["final_layout"] = gpu.LAYOUT_PRESENT
let rp = gpu.create_render_pass([attach])
let framebuffers = gpu.create_swapchain_framebuffers(rp)

let cmd_pool = gpu.create_command_pool()
let cmd = gpu.create_command_buffer(cmd_pool)
let img_sem = gpu.create_semaphore()
let rdr_sem = gpu.create_semaphore()
let fence = gpu.create_fence(true)

while gpu.window_should_close() == false:
    gpu.poll_events()
    gpu.wait_fence(fence)
    gpu.reset_fence(fence)
    let idx = gpu.acquire_next_image(img_sem)
    gpu.begin_commands(cmd)
    gpu.cmd_begin_render_pass(cmd, rp, framebuffers[idx], [[0.1, 0.1, 0.2, 1.0]])
    gpu.cmd_end_render_pass(cmd)
    gpu.end_commands(cmd)
    gpu.submit_with_sync(cmd, img_sem, rdr_sem, fence)
    gpu.present(idx, rdr_sem)

gpu.device_wait_idle()
gpu.shutdown_windowed()
```

## Quick Start: Triangle

```
import gpu

gpu.init_windowed("Triangle", 800, 600, "Sage Triangle", false)

# Load compiled SPIR-V shaders
let vert = gpu.load_shader("triangle.vert.spv", gpu.STAGE_VERTEX)
let frag = gpu.load_shader("triangle.frag.spv", gpu.STAGE_FRAGMENT)

# Render pass with swapchain format
let attach = {}
attach["format"] = gpu.FORMAT_SWAPCHAIN
attach["load_op"] = gpu.LOAD_CLEAR
attach["store_op"] = gpu.STORE_STORE
attach["initial_layout"] = gpu.LAYOUT_UNDEFINED
attach["final_layout"] = gpu.LAYOUT_PRESENT
let rp = gpu.create_render_pass([attach])

# Graphics pipeline
let layout = gpu.create_pipeline_layout([], 0)
let cfg = {}
cfg["layout"] = layout
cfg["render_pass"] = rp
cfg["vertex_shader"] = vert
cfg["fragment_shader"] = frag
cfg["topology"] = gpu.TOPO_TRIANGLE_LIST
cfg["cull_mode"] = gpu.CULL_NONE
let pipeline = gpu.create_graphics_pipeline(cfg)

# Render loop draws 3 hardcoded vertices
# gpu.cmd_bind_graphics_pipeline(cmd, pipeline)
# gpu.cmd_draw(cmd, 3, 1, 0, 0)
```

## GPU Module API Reference

### Context Lifecycle

| Function                                                       | Description                                          |
|----------------------------------------------------------------|------------------------------------------------------|
| <code>gpu.has_vulkan()</code>                                  | Returns <code>true</code> if Vulkan is available     |
| <code>gpu.has_window</code>                                    | <code>true</code> if GLFW windowed mode is available |
| <code>gpu.initialize(name, validation?)</code>                 | Initialize headless Vulkan context                   |
| <code>gpu.init_windowed(name, w, h, title, validation?)</code> | Initialize windowed Vulkan + GLFW + swapchain        |
| <code>gpu.shutdown()</code>                                    | Destroy headless context                             |
| <code>gpu.shutdown_windowed()</code>                           | Destroy window + context                             |
| <code>gpu.device_name()</code>                                 | GPU device name string                               |
| <code>gpu.device_limits()</code>                               | Dict of device limits                                |

### Windowing

| Function                                                      | Description                        |
|---------------------------------------------------------------|------------------------------------|
| <code>gpu.window_should_close()</code>                        | Check if window close requested    |
| <code>gpu.poll_events()</code>                                | Process window events              |
| <code>gpu.swapchain_extent()</code>                           | Dict {width, height}               |
| <code>gpu.swapchain_image_count()</code>                      | Number of swapchain images         |
| <code>gpu.acquire_next_image(semaphore)</code>                | Get next image index (-1 on error) |
| <code>gpu.present(image_index, wait_semaphore)</code>         | Present rendered image             |
| <code>gpu.create_swapchain_framebuffers(render_passes)</code> | Array of framebuffer handles       |

## Resources

| Function                                              | Description                       |
|-------------------------------------------------------|-----------------------------------|
| <code>gpu.create_buffer(size, usage, memory)</code>   | Create GPU buffer, returns handle |
| <code>gpu.buffer_upload(handle, float_array)</code>   | Upload data to buffer             |
| <code>gpu.buffer_download(handle)</code>              | Download data from buffer         |
| <code>gpu.create_image(w, h, d, format, usage)</code> | Create GPU image with auto view   |
| <code>gpu.create_sampler(mag, min, address)</code>    | Create texture sampler            |
| <code>gpu.load_shader(path, stage)</code>             | Load SPIR-V shader module         |

## Descriptors

| Function                                                               | Description                        |
|------------------------------------------------------------------------|------------------------------------|
| <code>gpu.create_descriptor_layout(bindings)</code>                    | Create from array of binding dicts |
| <code>gpu.create_descriptor_pool(max_sets, sizes)</code>               | Create descriptor pool             |
| <code>gpu.allocate_descriptor_set(pool, layout)</code>                 | Allocate a descriptor set          |
| <code>gpu.update_descriptor(set, binding, type, resource)</code>       | Bind buffer/image to descriptor    |
| <code>gpu.update_descriptor_image(set, binding, image, sampler)</code> | Bind combined image sampler        |

## Pipelines

| Function                                                              | Description                          |
|-----------------------------------------------------------------------|--------------------------------------|
| <code>gpu.create_pipeline_layout(layouts, push_size?, stages?)</code> | Create pipeline layout               |
| <code>gpu.create_compute_pipeline(layout, shader)</code>              | Create compute pipeline              |
| <code>gpu.create_graphics_pipeline(config_dict)</code>                | Create graphics pipeline (see below) |
| <code>gpu.create_render_pass(attachments)</code>                      | Create render pass                   |
| <code>gpu.create_framebuffer(render_pass, images, w, h)</code>        | Create framebuffer                   |

Graphics pipeline config dict keys: - `layout`, `render_pass`, `vertex_shader`, `fragment_shader` (required) - `topology` (default: `TOPO_TRIANGLE_LIST`) - `cull_mode`, `front_face`, `polygon_mode` - `depth_test`, `depth_write` (booleans) - `blend` (boolean, enables alpha blending) - `vertex_bindings`, `vertex_attribs` (arrays of dicts)

## Commands

| Function                                                                           | Description                  |
|------------------------------------------------------------------------------------|------------------------------|
| <code>gpu.create_command_pool()</code>                                             | Create command pool          |
| <code>gpu.create_command_buffer(pool)</code>                                       | Allocate command buffer      |
| <code>gpu.begin_commands(cmd)</code>                                               | Begin recording              |
| <code>gpu.end_commands(cmd)</code>                                                 | End recording                |
| <code>gpu.cmd_bind_compute_pipeline(cmd, pipe)</code>                              | Bind compute pipeline        |
| <code>gpu.cmd_bind_graphics_pipeline(cmd, pipe)</code>                             | Bind graphics pipeline       |
| <code>gpu.cmd_bind_descriptor_set(cmd, layout, index, set)</code>                  | Bind descriptor set          |
| <code>gpu.cmd_dispatch(cmd, x, y, z)</code>                                        | Dispatch compute workgroups  |
| <code>gpu.cmd_push_constants(cmd, layout, stage, data)</code>                      | Push constants (float array) |
| <code>gpu.cmd_begin_render_pass(cmd, rp, fb, clear)</code>                         | Begin render pass            |
| <code>gpu.cmd_end_render_pass(cmd)</code>                                          | End render pass              |
| <code>gpu.cmd_draw(cmd, verts, instances, first_v, first_i)</code>                 | Draw call                    |
| <code>gpu.cmd_draw_indexed(cmd, indices, instances, first, offset, first_i)</code> | Indexed draw                 |
| <code>gpu.cmd_bind_vertex_buffer(cmd, buffer)</code>                               | Bind vertex buffer           |
| <code>gpu.cmd_set_viewport(cmd, x, y, w, h, min_d, max_d)</code>                   | Set viewport                 |
| <code>gpu.cmd_set_scissor(cmd, x, y, w, h)</code>                                  | Set scissor rect             |
| <code>gpu.cmd_pipeline_barrier(cmd, src, dst, src_acc, dst_acc)</code>             | Memory barrier               |
| <code>gpu.cmd_image_barrier(cmd, img, old, new, src, dst, s_acc, d_acc)</code>     | Image barrier                |

## Synchronization

| Function                                                            | Description              |
|---------------------------------------------------------------------|--------------------------|
| <code>gpu.create_fence(signaled?)</code>                            | Create fence             |
| <code>gpu.wait_fence(fence, timeout?)</code>                        | Wait for fence           |
| <code>gpu.reset_fence(fence)</code>                                 | Reset fence              |
| <code>gpu.create_semaphore()</code>                                 | Create semaphore         |
| <code>gpu.submit(cmd, wait?, signal?, fence?)</code>                | Submit to graphics queue |
| <code>gpu.submit_compute(cmd, wait?, signal?, fence?)</code>        | Submit to compute queue  |
| <code>gpu.submit_with_sync(cmd, wait_sem, signal_sem, fence)</code> | Full sync submit         |
| <code>gpu.device_wait_idle()</code>                                 | Wait for GPU idle        |

| Function | Description |
|----------|-------------|
|----------|-------------|

## Key Constants

# Buffer usage (bitwise OR)

```
gpu.BUFFER_STORAGE  gpu.BUFFER_UNIFORM  gpu.BUFFER_VERTEX
gpu.BUFFER_INDEX    gpu.BUFFER_STAGING  gpu.BUFFER_TRANSFER_SRC
```

# Memory

```
gpu.MEMORY_DEVICE_LOCAL  gpu.MEMORY_HOST_VISIBLE  gpu.MEMORY_HOST_COHERENT
```

# Formats

```
gpu.FORMAT_RGBA8      gpu.FORMAT_RGBA16F  gpu.FORMAT_RGBA32F
gpu.FORMAT_R32F       gpu.FORMAT_DEPTH32F  gpu.FORMAT_SWAPCHAIN
```

# Shader stages

```
gpu.STAGE_VERTEX  gpu.STAGE_FRAGMENT  gpu.STAGE_COMPUTE  gpu.STAGE_ALL
```

# Descriptor types

```
gpu.DESC_STORAGE_BUFFER  gpu.DESC_UNIFORM_BUFFER
gpu.DESC_SAMPLED_IMAGE   gpu.DESC_STORAGE_IMAGE  gpu.DESC_COMBINED_SAMPLER
```

# Topology

```
gpu.TOPO_TRIANGLE_LIST  gpu.TOPO_LINE_LIST  gpu.TOPO_POINT_LIST
```

# Pipeline stages (barriers)

```
gpu.PIPE_TOP  gpu.PIPE_COMPUTE  gpu.PIPE_TRANSFER  gpu.PIPE_FRAGMENT
```

# Access flags (barriers)

```
gpu.ACCESS_SHADER_READ  gpu.ACCESS_SHADER_WRITE
gpu.ACCESS_TRANSFER_READ  gpu.ACCESS_HOST_READ
```

## Demos

Three demo programs are included in `examples/`:

| Demo           | File                                    | Description                                              |
|----------------|-----------------------------------------|----------------------------------------------------------|
| Empty Window   | <code>examples/gpu_window.sage</code>   | Creates a window with cycling clear color                |
| Triangle       | <code>examples/gpu_triangle.sage</code> | Basic colored triangle via vertex/fragment shaders       |
| 3D Hello World | <code>examples/gpu_hello3d.sage</code>  | Rotating "HELLO WORLD" text rendered as 3D line segments |

Run them with:

```
./sage examples/gpu_window.sage
./sage examples/gpu_triangle.sage
./sage examples/gpu_hello3d.sage
```

Shaders are pre-compiled SPIR-V files in `examples/shaders/`. Recompile with:

```
cd examples/shaders
glslc triangle.vert -o triangle.vert.spv
glslc triangle.frag -o triangle.frag.spv
```

```
glslc text3d.vert -o text3d.vert.spv
glslc text3d.frag -o text3d.frag.spv
```

---

## Appendix: Quick Reference

### Keywords

and as async await break case catch class comptime continue default defer elif else end enum false final

*Note: `print`, `end`, `match`, `init`, `enum`, `struct`, and `trait` are soft keywords, meaning they can be used as variable, property, or method names where unambiguous.*

### Built-in Functions

```
len(x) push(arr, val) pop(arr) append(arr, val) range(a, b)
split(str, delim) join(arr, sep) replace(s, old, new)
upper(s) lower(s) strip(s) slice(arr, a, b)
str(x) tonumber(s) int(x) input() clock()
type(x) chr(n) ord(c) hash(x) sizeof(x) doc(x)
startswith(s, prefix) endswith(s, suffix)
contains(s, sub) indexof(s, sub) string_count(s, sub) string_repeat(s, n)
array_extend(a, b) array_repeat(a, n) array_reverse(a)
array_contains(a, v) array_index_of(a, v) array_min(a) array_max(a)
array_sum(a) array_product(a)
bytes() bytes_len(b) bytes_get(b, i) bytes_set(b, i, v)
bytes_push(b, v) bytes_slice(b, s, e) bytes_to_string(b)
dict_keys(d) dict_values(d) dict_has(d, k) dict_delete(d, k)
gc_collect() gc_stats() gc_collections() gc_enable() gc_disable()
gc_mode() gc_set_arc() gc_set_orc()
next(gen)
ffi_open(path) ffi_call(lib, fn, ret, ...) ffi_sym(lib, name) ffi_close(lib)
ffi_sym_addr(lib, name)
mem_alloc(n) mem_free(ptr) mem_read(ptr, off, type) mem_write(ptr, off, type, val)
mem_size(ptr) addressof(val) addressof_raw(val) ptr_add(ptr, n) ptr_to_int(ptr)
path_join(a, b) path_dirname(p) path_basename(p) path_ext(p)
path_exists(p) path_is_dir(p) path_is_file(p)
asm_exec(code, ret, ...) asm_compile(code, arch, out) asm_arch()
struct_def(fields) struct_new(def) struct_get(ptr, def, name)
struct_set(ptr, def, name, val) struct_size(def)
cpu_count() cpu_physical_cores() cpu_has_hyperthreading()
thread_set_affinity(id) thread_get_core()
atomic_new(v) atomic_load(a) atomic_store(a, v)
atomic_add(a, v) atomic_cas(a, e, d) atomic_exchange(a, v)
sem_new(n) sem_wait(s) sem_post(s) sem_trywait(s)
vm_gas_limit_set(n) vm_gas_used_get() vm_gas_limit_get()
build_quad_verts(quads) build_line_quads(lines, thickness, r, g, b, a)
```

### Operators

```
Arithmetic:    + - * / %
Comparison:    == != < > <= >=
Logical:       and or not
Bitwise:       & | ^ ~ << >>
Assignment:    =
```

|           |                                                      |
|-----------|------------------------------------------------------|
| Indexing: | <code>arr[i]</code>                                  |
| Slicing:  | <code>arr[a:b]</code>                                |
| Property: | <code>obj.field</code> or <code>obj-&gt;field</code> |
| Call:     | <code>func(args)</code>                              |